Estimating Poverty Incidence in the Philippines

Using Satellite Imagery and CNN–LSTM Architectures

by

Aldave, Ruben Jr. V.
Grün, David D.
Palma, Ralph Christian M.
Viacrusis, Sean Matthew L.

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in Computer Science with Specialization in Data Science

at

FEU Institute of Technology

May 2026

Justine Jude C. Pura
Thesis Adviser

APPROVAL AND ACCEPTANCE SHEET

The thesis entitled **"Estimating Poverty Incidence in the Philippines Using Satellite Imagery and CNN–LSTM Architectures"** prepared and submitted by:

Aldave, Ruben Jr. V.
Grün, David D.
Palma, Ralph Christian M.
Viacrusis, Sean Matthew L.

In partial fulfillment of the course of requirement for the Degree of Bachelor of Science in Computer Science with specialization in Data Science has been examined and is hereby recommended for approval.

MS. ELISA V. MALASAGA
Panelist 1

DR. DENNIS B. GONZALES
Panelist 2

MS. FANNY C. ALMENIANA
Head Panelist

Accepted as partial fulfillment of the requirements for the **Degree of Bachelor of Science in Computer Science with specialization in Data Science**.

JUSTINE JUDE C. PURA
Thesis Adviser

JUSTINE JUDE C. PURA
Thesis Mentor

DR. SHANETH C. AMBAT
Program Director

May 2026

ABSTRACT

In the Philippines, poverty remains a hindrance to the development of the nation. While overall poverty rates fell from 9.9% since 2000 to 6.2% by 2018, this development has largely been unequal, with underdeveloped regions in the country still being left behind with more than 60% of the population still living below the poverty line. There is, therefore, a need to identify and map the most disadvantaged segments of the population in order to target and address their needs. In the Philippines, this is done by conventional means such as the Family Income and Expenditure Survey and the Demographic and Health Survey. While these offer detailed and comprehensive snapshots of poverty and other demographic characteristics in the country, these surveys are labor and resource-intensive and and time-consuming; consequently, they are conducted infrequently. With the advancements in artificial intelligence and machine learning and the rise of unconventional sources of data such as satellite imagery, social media data, and crowd sourced spatial data, technology can significantly complement poverty mapping and distribution. This study, therefore, aims to use satellite imagery and poverty and demographic data to generate geographic poverty estimates in select localities in the Philippines.

Keywords: Poverty, Satellite Imagery, Remote Sensing, Demographic, Machine Learning, Geospatial Artificial Intelligence

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INTRODUCTION

The proposed study investigates the use of advanced deep learning techniques to map and estimate poverty levels from remotely sensed data. Traditional poverty measurement methods in the Philippines often rely on time-consuming and resource-intensive household surveys, which can be limited in frequency and geographic coverage. Recent studies have demonstrated that machine learning can associate and predict poverty and socioeconomic indicators from satellite images. This study integrates a Convolutional Neural Network (CNN) to extract spatial and textural features from satellite images and a Long Short-Term Memory (LSTM) to model temporal patterns in socioeconomic changes within the Philippine setting, an approach that has yet to be undertaken in this context. This research aims to capture both spatial and temporal dimensions of poverty to produce more accurate and timely poverty maps that can support policy-making and targeted interventions.

1.1 Background of the Study

Poverty remains a persistent barrier to the Philippines' development, marked by stark urban-rural divides. While national poverty incidence was 15.5% in 2023, regional disparities are extreme: the National Capital Region reported only 1.1%, whereas some rural regions exceed 60% (Onsay & Rabajante, 2024). This inequality extends beyond income to overcrowding in urban slums and lack of basic services in remote provinces. For policymakers, this spatial heterogeneity makes "one-size-fits-all" approaches ineffective, as urban and rural poverty differ fundamentally in nature.

The Philippine government has launched initiatives like AmBisyon Natin 2040, which envisions ending poverty (National Economic and Development Authority, 2015). Central to these efforts are social protection programs like the Pantawid Pamilyang Pilipino Program (4Ps) and digital dashboards such as E-GOV PH at the barangay level (Department of Information and Communications Technology, 2019). However, these dashboards face technical constraints—limited rural internet connectivity and low digital proficiency—resulting in fragmented or politicized data that hinders resource allocation based on objective, real-time needs (Jou et al., 2024).

To bridge this information gap, the government traditionally relies on the Family Income and Expenditure Survey (FIES) and the Demographic and Health Survey (DHS). These instruments are considered the gold standard for poverty mapping in the Philippines, providing high-quality, granular insights into household assets and consumption patterns (Philippine Statistics Authority & ICF, 2023). By analyzing these datasets, NEDA and the PSA can identify which provinces are falling behind and adjust macroeconomic management accordingly. These surveys are essential for evidence-based governance, as they allow for the calculation of the poverty threshold in terms of wealth index, the minimum income required for a family to meet basic food and non-food needs (Onsay & Rabajante, 2024).

However, traditional surveying methods possess significant disadvantages that limit their utility for agile decision-making. The primary drawback is their infrequency and cost; a single national survey like the DHS can cost between $1.5 million and $2 million USD and takes years to complete, from data collection to final publication (Burke et al., 2021). This data lag means that by the time policymakers receive poverty estimates, the economic conditions on the ground which are often affected by sudden shocks like typhoons, conflicts or global inflation may have already shifted, rendering the data outdated and leading to unavoidable inclusion and exclusion errors in policy decisions, grant allocations, and program proposals.

The emergence of unconventional data sources, specifically utilizing daytime multispectral imagery and nighttime luminosity data from satellite sensors, offers an innovative solution to these challenges. By utilizing CNN-LSTM (Convolutional Neural Network – Long Short-Term Memory) architectures, researchers can extract spatial features from satellite imagery and model patterns that traditional surveys miss. Machine learning models can predict economic well-being at a high frequency and high resolution, acting as an impact multiplier for ground-based data collection (Burke et al., 2021). For Philippine decision-makers, integrating satellite-based poverty estimates into their workflows provides the real-time insights necessary to target the most disadvantaged segments of the population accurately, ensuring that no community is left behind in the pursuit of national development.

1.2 Significance of the Study

This study aims to provide a modular, data-driven framework for poverty estimation and spatial analysis using a hybrid Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) architecture, producing poverty map visualizations that enable evidence-based decision-making. The proposed system supports more efficient identification of high-poverty areas, improved targeting of social protection programs, and data-driven resource allocation, complementing existing survey-based approaches. The following stakeholders stand to benefit most from this study:

- **Philippine government agencies and policymakers.** Particularly the Department of Social Welfare and Development (DSWD), and the Department of Economy, Planning, and Development (DEPDev) as the study will enable these institutions to identify geographic areas requiring prioritized intervention, support the targeting of social protection programs such as the Pantawid Pamilyang Pilipino Program (4Ps), and provide evidence-based justification for budget allocation and poverty reduction proposals.

- **Non-government organizations (NGOs) and humanitarian aid groups.** By showcasing reliable, spatiotemporal visualizations of poverty clusters, this study empowers NGOs to identify underserved and vulnerable communities that may be overlooked by broad national surveys. This system provides independent, data-driven evidence to validate funding proposals and optimize the distribution of aid. This ensures that limited resources, such as food security programs, educational initiatives, and disaster relief are directed toward the most marginalized populations with greater precision and impact.

- **Academic researchers and institutions.** The study contributes to the body of knowledge in remote sensing and geospatial artificial intelligence (GeoAI) by demonstrating the application of a many-to-one LSTM architecture within the Philippine context. It serves as a methodological precedent for handling tropical constraints such as cloud cover and seasonal agricultural changes in socioeconomic analysis. This research will serve as a foundational reference for future studies seeking to integrate multisource geospatial data, deep learning, and social science datasets to address complex societal challenges.

1.3 Statement of the Problem

The persistent spatial inequality in the Philippines creates prolonged delays in humanitarian intervention and resource misallocation for marginalized regional populations. Despite the data provided by government agencies such as the Philippine Demographic and Health Survey, the absence of high-frequency, scalable monitoring tools makes it difficult to maintain up-to-date poverty maps between census cycles. To address this gap, this study aims to develop a spatial poverty estimation framework, utilizing a hybrid CNN-LSTM architecture to provide granular, near-real-time socioeconomic estimates from satellite imagery. Specifically, this study seeks to answer the following questions:

1. How can heterogeneous data sources, specifically daytime and nighttime satellite imagery, spatial features data, and socioeconomic survey data, be collected and preprocessed to create a cohesive and reliable dataset for model training?

2. How can transfer learning be applied to a Convolutional Neural Network (CNN) using nighttime luminosity as a proxy task to enable the extraction of robust spatial features from daytime satellite imagery?

3. How can static infrastructure features and dynamic spatiotemporal features be effectively fused within a hybrid neural network architecture to improve the

regression of household wealth?

4. What is the predictive performance of the combined model when assessed using standard regression metrics, including Root Mean Squared Error (RMSE), Mean Squared Error (MSE), the Pearson Correlation Coefficient (r), and the Coefficient of Determination (R^2)?

5. How can the hybrid neural network's estimates be interpreted to quantify the relative importance of different spatial and temporal features?

6. How can the estimated poverty levels be translated into a user-facing visualization module to facilitate data interpretation for policy analysis?

1.4 Objectives

The general objective of this study is to develop a system capable of estimating poverty levels across the Philippines using satellite-derived data and deep learning techniques. In particular, this study aims to:

- collect and preprocess satellite imagery from Sentinel 2 and Visible Infrared Imaging Radiometer Suite (VIIRS), spatial features and points of interest data from OpenStreetMap via Geofabrik, and corresponding socioeconomic data from the 2022 Demographic and Health Survey to create a reliable dataset for model training and validation, and additionally collect full spatial coverage imagery for select localities in the Philippines for output generation;

- implement a transfer learning strategy by pre-training a Convolutional Neural

Network (CNN) on a nighttime light classification proxy task to enable the model to learn latent representations of economic development from daytime imagery;

- develop a hybrid regression architecture that processes temporal satellite features using a Long Short-Term Memory (LSTM) network and fuses them with static infrastructure indices such as road density, building counts, POIs derived from OpenStreetMap to estimate the DHS Wealth Index;
- evaluate and validate the model's predictive performance using standard metrics such as Mean Squared Error (MSE) and Coefficient of Determination (R^2);
- implement SHAP (SHapley Additive exPlanations) value analysis as an interpretability method to quantify the marginal contribution of each input feature to the model's estimated DHS Wealth Index, thereby providing transparent and auditable explanations of the model's behavior; and
- develop a visualization module that shows the estimated poverty levels using heat maps and analysis reports for interpretation and policy use.

1.5 Scope and Delimitations

1.5.1 Scope of the Study

This study focuses on the development of a spatiotemporal deep learning system designed to estimate poverty levels across the entire Philippine archipelago. The model training is grounded in ground-truth socioeconomic data from the 2022 Demographic and Health Survey (DHS), utilizing all available survey clusters nationwide to maximize the poverty gradient representation. The generation of poverty maps and SHAP-based analytics reports, however, is scoped to eleven local governmental units: the five provinces with the highest poverty incidence, the five provinces with the lowest poverty incidence as shown in table 1, and the National Capital Region. This deliberate contrast between the poorest and wealthiest provinces maximizes the analytical value of the model, enabling direct comparison of the spatial and infrastructural drivers of poverty across opposite ends of the socioeconomic spectrum.

The system integrates three primary data modalities: optical daytime imagery from Sentinel-2 and nighttime luminosity data from the Visible Infrared Imaging Radiometer Suite (VIIRS), both acquired in collaboration with the Philippine Space Agency and processed into quarterly median composites to enable spatiotemporal analysis; static crowdsourced geospatial features, specifically points of interest (POI), road networks, and buildings from OpenStreetMap (OSM); and the DHS Wealth Index, which serves as the operational definition of poverty. Methodologically, the study implements a late fusion CNN-LSTM architecture, which combines temporal satellite patterns with static geospatial context to regress the DHS Wealth Index. The study further includes an interpretability layer implemented through SHAP value analysis, which is applied to the trained hybrid model to generate localized analytics reports. These reports are integrated into the visualization module and are made available for download by end-users to support area-specific policy analysis.

Province	Poverty Incidence Among Families (%)	Province	Poverty Incidence Among Families (%)
Ilocos Norte	0.3	Zamboanga del Norte	37.7
Pampanga	1.0	Basilan	33.7
Benguet	2.5	Tawi-tawi	32.3
Kalinga	2.6	Maguindanao del Sur	32.1
Aklan	3.1	Davao Oriental	29.1

Table 1: Selected provinces included in the study. Data from Philippine Statistics Authority (2023)

1.5.2 Delimitations of the Study

The study operates under several technical and data constraints. First, the system relies on 10-meter satellite imagery resolution as a proxy for poverty. Second, while the input satellite data is temporal, the ground-truth DHS data is static; thus, the study assumes that the socioeconomic status of a cluster remained relatively stable throughout the observation year. Likewise, wealth index predictions for succeeding years represent projected estimates based on visual indicators rather than validated ground truth. Another significant geospatial limitation is the inherent random displacement of DHS GPS coordinates, introduced to protect respondent anonymity, which necessitates the use of cluster-buffer methods that introduce a margin of spatial uncertainty. Furthermore, despite the use of quarterly median compositing to mitigate noise, persistent cloud cover in tropical regions may still result in data gaps. Inference for other localities in the Philippines, while technically feasible with the trained model, is outside the scope of the current study and is identified as a direction for future research. Finally, the study is restricted to model training, validation, and the generation of visualization maps, and does not extend to the development of a real-time monitoring system due to the inherent latency of satellite data acquisition and processing.

1.6 Theoretical Framework

1.6.1 Utilization of Satellite Imagery

The theoretical basis for utilizing satellite imagery in this study addresses the structural limitations of traditional poverty estimation methods, such as the Family Income and Expenditure Survey (FIES) and the Demographic and Health Survey (DHS). While these surveys serve as the standard for poverty mapping, their utility is constrained by high costs and infrequency, which often results in significant data lags that fail to capture immediate economic shifts caused by external shocks such as global and economic crises. To bridge these gaps, this study adopts the theoretical perspective that physical environmental features present in satellite images serve as reliable alternatives for estimating poverty. This is supported by the study of Martinez et al. (2016) stating that nighttime luminosity acts as a significant indicator of wealth and poverty levels, especially in remote areas that are not easily reached by census workers. Their findings validate that differences in nighttime lights could directly correlate with electrification rates and economic activity, providing an alternative to traditional surveys.

Furthermore, the framework suggests that satellite imagery functions as an impact multiplier rather than a replacement for traditional data collection. As stated by Burke et al. (2021), machine learning models can predict economic status at high frequency and high resolution when calibrated with traditional survey data. This concept is supported locally by Philippine Institute for Development Studies (2020), who cited that combining cost-efficient machine learning models with accessible geospatial information offers a low-cost, scalable method for determining poverty estimates. This theoretical alignment allows the study to leverage the accessibility of satellite data to bridge data gaps and provide policymakers with granular, up-to-date insights necessary to target rural areas accurately (Philippine Institute for Development Studies, 2020).

1.6.2 Deep Learning Models

The computational framework for interpreting satellite data is anchored on Deep Learning, specifically the ability of algorithms to automatically identify patterns in complex images. For the spatial analysis component, the study utilizes Convolutional Neural Networks (CNNs), theoretically based on the work of LeCun et al. (2015). They defined deep learning as a set of techniques that allow computational models to learn representations of data with multiple levels of abstraction, which is crucial for handling complex datasets like high-resolution satellite imagery.In the context of poverty mapping, Yang et al. (2022) demonstrated that CNNs are highly effective for this task as they can independently recognize complicated structures. This capability allows the model to detect physical features from satellite imagery such as infrastructure, building density, and road networks which serves as significant indicators of household wealth.

To address the temporal aspects of poverty, the framework incorporates Long Short-Term Memory (LSTM) networks which are based on Recurrent Neural Network (RNN). LSTMs are specifically made to solve the vanishing gradient problem in sequence modeling, which enables them to capture non-linear patterns and temporal dependencies in time-series data, as confirmed by Absar et al. (2022). The system's ability to analyze historical sequences and differentiate between steady economic activity and transient anomalies is made possible by this theoretical capability, which is essential for this study. Lastly, the study of Srishti Gulecha et al. (2024) support the combination of these two models into a Hybrid CNN-LSTM Architecture. Their results demonstrate that, in comparison to single-model approaches, combining temporal trends via LSTM with spatial features via CNN results in noticeably higher predictive accuracy. This theoretically justifies the hybrid methodology as the most robust approach for capturing the dynamic and multifaceted nature of poverty in the Philippines.

1.7 Conceptual Framework

Figure 1 outlines the integration of satellite-derived spatial data and survey-based socioeconomic data through hybrid deep learning methods for model development, validation, and expert-guided evaluation.

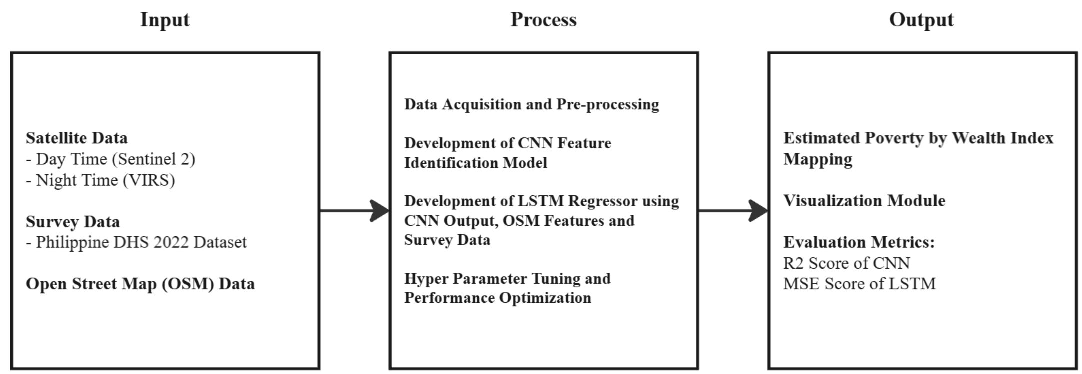

Figure 1: Conceptual framework

1.8 Definition of Terms

CNN–LSTM Hybrid Model – An integrated deep learning architecture that combines a Convolutional Neural Network (CNN) for spatial feature extraction from satellite imagery and a Long Short-Term Memory (LSTM) network for modeling temporal socioeconomic patterns.

Convolutional Neural Network (CNN) – A deep learning model designed to automatically extract spatial and textural features from image data, used in this study to analyze satellite imagery related to poverty indicators.

Data Preprocessing – The set of procedures applied to raw satellite and survey data, including cleaning, normalization, alignment, and transformation, prior to model training and analysis.

Dashboard Interface – A visualization module developed in this study to display poverty distribution maps and temporal trends for expert assessment and policy interpretation.

Long Short-Term Memory (LSTM) – A type of recurrent neural network capable of capturing temporal dependencies in sequential data, used in this study to model socioeconomic changes over time.

Model Validation – The process of evaluating the accuracy, reliability, and generalizability of the proposed model using unseen or test data.

Poverty – A socioeconomic condition characterized by limited access to basic needs, services, and opportunities, as operationalized in this study through spatial and survey-based indicators.

Poverty Incidence – The proportion of households in a given cluster or locality living below the poverty threshold, as defined by the DHS Wealth Index in this study.

Poverty Mapping – The process of estimating and visualizing the spatial distribution of poverty levels using geospatial data and computational modeling techniques.

Remote Sensing – The acquisition of information about the Earth's surface without direct physical contact, primarily through satellite-based sensors.

Satellite Imagery – Remotely sensed images of the Earth's surface obtained from satellite platforms, used in this study to extract spatial features associated with socioeconomic conditions.

SHAP Values (SHapley Additive exPlanations) – A model-agnostic post-hoc interpretability method grounded in cooperative game theory, specifically Shapley values. SHAP values are the analytical basis for generating localized analytics reports.

Socioeconomic Data – Survey-based information describing household characteristics, population attributes, and living conditions, sourced from the Philippine

Demographic and Health Survey (DHS) and the Philippine Statistics Authority (PSA).

Spatial Features – Physical and environmental characteristics derived from satellite imagery, such as infrastructure density, vegetation coverage, and urban development patterns.

Temporal Patterns – Trends or changes in socioeconomic indicators over time, captured in this study through sequential modeling techniques.

Wealth Index –A composite indicator of household socioeconomic status derived using principal component analysis (PCA) of asset ownership, housing characteristics, household composition, and water and sanitation variables.

Chapter 2

REVIEW OF RELATED LITERATURE

2.1 Local Literature

Every three years, the Family Income and Expenditure Survey (FIES) is conducted to gather detailed data on a family's income and expenditures. The FIES is undertaken to determine the degree of inequality through income distribution and to estimate poverty incidence. In the 2025 FIES, the PSA, through the Social Sector Statistics Service, has been given a budget allocation of 571 million Philippine Pesos to conduct the nationwide survey. Results are expected to be released in August 2026 (Philippine Statistics Authority, 2025). Other traditional surveying methods or initiatives include the Community-Based Monitoring System institutionalized by the Republic Act No. 11315 and the Listahanan, commandeered by the Department of Social Welfare and Development (DSWD).

In light of that, Reyes et al. (2010) stated in a discussion paper that there are insufficient poverty reduction programs. The population continues to grow, and more challenging events such as economic crises, natural disasters, and even pandemics carry on. While said strategies have not been established yet, people falling into poverty in times of economic crises, natural disasters, and life-cycle shocks go undocumented. It was cited that there are significant movements in and out of poverty, highlighting the vulnerabilities of some parts of the population to unanticipated events.

It is stated by the Philippine Institute for Development Studies (2020) that satellite imagery and other unconventional data sources can be a significant alternative for fast and inexpensive estimation of the country's socioeconomic indicators. Thinking Machines Data Science Inc. Founder and Chief Executive Officer Stephanie Sy stated that combining a cost-efficient machine learning model with accessible geospatial information could be a low-cost, scalable, and fast method for determining poverty estimates. In addition, it was mentioned that technology could be used to gather useful data where surveys are not feasible. However, Sy also emphasized that traditional surveys and machine learning models should be complementary, not replacements for each other.

Vizmanos et al. (2022) discussed that the utilization of new data sources, such as big data and crowd-sourced data, could be significantly beneficial in complementing traditional sources of statistics and uncovering more insights. To further explain, it is stated that the examination of these new data sources could assist in developing interventions through policies and actions that focus on attaining sustainable and inclusive development. It was also mentioned that the Philippines continues to have data gaps for monitoring the country's development and global goals, which signifies the need for new data sources, such as big data and several more. It was also cited that integrating satellite imagery with census and survey data significantly boosted the quality of poverty statistics in smaller areas in Thailand.

Furthermore, the value of using non-traditional data is further supported by the successful use of satellite technology to track local economies. Martinez et al. (2016) stated that the nighttime luminosity is a reliable indicator of a province's wealth and poverty levels in the Philippines. Additionally, it is stated to be useful for reaching remote areas where it is difficult for census workers to travel. Detailed poverty maps could be created by simply looking at physical details like the number of buildings and the intensity of streetlights. This suggests that the utilization of non-conventional data sources, such as satellite imagery, offers a faster, more flexible way to monitor poverty.

2.2 Foreign Literature

Hall et al. (2023) cited in a study the increasing use case for LSTM networks in further enhancing already significant predictive results achieved by traditional neural networks by capturing temporal predictions without suffering optimization hurdles. This was further emphasized by mentioning a study by Yang et al. (2022) that used a hybrid approach, specifically the CNN-LSTM model to achieve improved prediction results for surface erosions rates with significantly reduced time and computational costs. Thus, the researchers recommended more studies to adopt the hybrid model, CNN-LSTM, for poverty-satellite data ML domain of research. This suggests that the high-resolution spatial feature extraction provided by CNNs, when paired with the temporal sequence learning of LSTMs, could produce cost-effective and high-frequency poverty estimates.

Hall et al. (2023) also emphasized that machine learning algorithms combined with satellite imagery have revolutionized poverty prediction, offering critical implications for development research. The results of the study also found that the combination of machine learning and deep learning algorithms could significantly improve predictive power by 15% compared to using them alone. This technological shift is further supported by Lamichhane et al. (2025), who highlighted the increasing reliability of remote sensing data in identifying socioeconomic patterns across various global contexts. It was mentioned in the study that supervised and unsupervised machine learning techniques are widely used for poverty research. The researchers also noted that deep learning and gradient boosting may deliver better results and accuracy. Most importantly, the researchers recommended exploring studies incorporating both traditional and technological models for improved performance.

The novelty of these mapping techniques also lies in the advancements of deep learning algorithms. As described in a study by LeCun et al. (2015), deep learning is a powerful set of techniques for learning in neural networks that allows a computational model of multiple processing layers to learn representations of data with multiple levels of abstraction. This ability is crucial, especially when dealing with complex data sets, such as high-resolution satellite images for which the system has to independently learn the intricate structures necessary for making reliable predictions.

In a study by Yang et al. (2022), researchers used Convolutional Neural Networks (CNNs) to handle spatial features. It is designed to search for important, complicated patterns from high-dimensional data. This network works well on data presented in multiple arrays and can discover complex structures by itself. In the context of poverty mapping, this enables the detection of physical features such as infrastructure, building density, and light intensity directly from raw spatial data.

Absar et al. (2022) used a deep learning-based Long Short-Term Memory (LSTM) model to predict the daily course of infectious disease outbreaks based on sequences of historical infection data in order to estimate cases ahead. The model being used was developed in order to address the temporal evolution of the health crisis through temporal regression, which enabled researchers to find complex and non-linear patterns, including time-sequential data, as well as produce precise forecasts on how an outbreak would progress. This methodological technique highlights the ability of LSTM in learning from historical sequences to fill critical gaps in data during dynamically evolving events.

2.3 Synthesis of Related Literature

The review of both foreign and local related literature reveals a significant challenge in the current methods of poverty estimation or monitoring in the Philippines. The Family Income and Expenditure Survey (FIES) remains as the primary instrument for measuring poverty in the Philippines. However, the standard three-year interval between the results of the said surveys poses difficulties in capturing the immediate effects of shocks such as natural disasters, economic downturns, and public health crises. These shocks could result in people falling into poverty (Philippine Statistics Authority, 2025; Reyes et al., 2010).

To address data gaps, researchers like Vizmanos et al. (2022) and Philippine Institute for Development Studies (2020) are recommending for the utilization of crowd-sourced data and big data such as satellite imagery to determine poverty estimates and bridge data gaps in monitoring the country's development. In addition, Martinez et al. (2016) stated that nighttime luminosity and building density could further assist in estimating wealth in remote areas that census workers cannot easily reach. An approach that combines satellite imagery and machine learning models thus provides a cost-effective, scalable platform to compensate for the logistical constraints of regular surveys, bridging significant data gaps and allowing more granular and timely insight into socio-economic indicators (Martinez et al., 2016; Philippine Institute for Development Studies, 2020).

Furthermore, the technical feasibility of the use of satellite imagery and machine learning is supported by several foreign literature which indicates that combining machine learning with remote sensing can enhance predictive accuracy up to 15% (Hall et al., 2023; Lamichhane et al., 2025). This is further enhanced through the use of deep learning models that could learn to see patterns in raw data that people might miss, ensuring more reliable predictions (LeCun et al., 2015). To be specific, Convolutional Neural Networks (CNNs) are effective at identifying spatial indicators such as building density, light intensity, and infrastructure (Yang et al., 2022). On the other hand, the Long Short-Term Memory (LSTM) models offer an alternative in analyzing historical sequence data which could be beneficial in filling critical gaps in data (Absar et al., 2022). Collectively, these findings validate the potential of a hybrid model to bridge critical data gaps, enhance predictive accuracy, and provide a scalable solution that effectively complements traditional surveying methods with real-time geospatial insights.

2.4 Local Studies

Recent research by the Asian Development Bank (Asian Development Bank, 2021) has demonstrated the potential of geospatial technologies in understanding poverty patterns in the Philippines. To supplement infrequent national census data and support evidence-based policymaking, the study aimed to generate granular, municipality-level poverty estimates. The researchers integrated multi-source geospatial data, including daytime satellite images from Google Earth, nighttime luminosity from VIIRS, OpenStreetMap (OSM) points-of-interest, and ground-truth data from the 2015 Family Income and Expenditure Survey (FIES). Methodologically, the study employed a transfer learning approach, utilizing a Convolutional Neural Network (CNN) for spatial feature extraction, which was then combined with Ridge Regression and Random Forest algorithms. This hybrid model achieved a R score of 0.63. Furthermore, the spatial analysis revealed that areas with fewer infrastructure facilities and dispersed dwellings exhibited higher poverty levels, underscoring the necessity of spatial data for targeting social programs. This study provides a crucial foundation for the present research by proving that satellite imagery is a reliable proxy for socioeconomic evaluation, directly validating this study's proposed use of CNNs for spatial feature extraction.

Building on the use of alternative data sources, a study by Aragoza and Reyes (2022) explored the application of machine learning algorithms in predicting the poverty situation in the Philippines by utilizing publicly available data, including nightlight intensity, points of interest, and internet speed. Different regression and classification techniques with the help of these features were used in combination with regional socioeconomic data. It was identified in the research that nightlight intensity and internet speed could effectively reflect the level of economic activities in the regions and thus can be used to reliably measure poverty incidence. The thesis is backed up by this work of research, which shows that nontraditional indicators may be helpful in making better poverty predictions if they are integrated with machine learning techniques.

Similarly, Tingzon et al. (2019) used satellite imagery and crowdsourced geospatial data to create higher-resolution poverty maps in rural regions. The method uses spatial analysis of field data on settlement patterns and infrastructure, which they then fed into machine learning models to predict poverty levels. To be specific, the researchers used data sources such as Google Static maps for daytime satellite images, VIIRS for nighttime images, OpenStreetMap data and a high resolution settlement data. They used deep learning models, specifically a CNN-Ridge hybrid model which got a total r-squared score of 0.63. Results indicated that the accuracy of prediction is much higher based on multi-source information than based strictly on survey data, particularly in poorly supported rural areas. This reinforces the thesis goal of harnessing the concept of utilizing multiple data sources and learning algorithms for improved poverty measurement.

In addition, a recent local study focused on combining free satellite images and machine learning to produce very detailed poverty estimates in the Philippines. The research used CNN models to identify spatial features in Sentinel and Google Earth images, and then these were correlated with the wealth indices coming from the survey data. The results showed that data obtained from remote sensing could be used to complement the traditional survey, thus allowing more frequent and more detailed geographically speaking poverty estimates. This research serves the CNN-LSTM method proposal as a direct reference since it has been shown that spatial feature extraction is quite effective in the context of the Philippines (Jimenez, 2025).

In contrast to methods relying heavily on automated neural network feature extraction, Ledesma et al. (2020) explored an interpretable, cost-efficient approach to poverty estimation using alternative, readily accessible data. The study aimed to predict asset wealth indices without the computational expense of high-resolution satellite imagery. To achieve this, the researchers fused low-resolution satellite data with crowdsourced OpenStreetMap (OSM) features and aggregated Facebook social media data, such as device types and network connectivity metrics. Methodologically, rather than utilizing Convolutional Neural Networks, the study engineered specific spatial and social features to feed into traditional machine learning regressors, specifically XGBoost and Random Forest. This multi-source methodology achieved an r-squared score of 0.66, successfully outperforming baseline models that relied exclusively on satellite data. While this research highlights the significant predictive power of integrating OSM and diverse alternative datasets, it also underscores the limitations of static regressors. Consequently, this justifies the implementation of a more advanced deep learning architecture capable of not only matching this predictive accuracy but also extracting complex spatial hierarchies and temporal trends that manual feature engineering may overlook.

2.5 Foreign Studies

Extending beyond the local context, several studies have investigated the use of deep learning for poverty mapping. Srishti Gulecha et al. (2024) aimed at identifying poverty areas in India by using machine learning and deep learning methods. Several CNN models were trained on high resolution satellite images to identify man made features like the density of buildings and urban facilities. At the same time, socio-economic data collected over time were integrated with LSTM networks. The outcomes revealed that a combination of spatial and temporal representation led to a higher level of prediction, which underlines the pertinence of CNN-LSTM frameworks when mapping poverty changes. This research supports the notion that hybrid deep learning approaches can be successfully applied in the context of developing nations where poverty is unevenly spread (Srishti Gulecha et al., 2024).

In a similar study by Saeed and Turkoglu (2023), convolutional neural networks (CNNs) were used to analyze satellite images for their poverty predictions. They specifically targeted rural areas where survey data was scarce. The methodology consists of preprocessing high-resolution images, employing CNN layers for feature extraction, and using regression to predict poverty indices. Their study showed that CNNs can accurately detect local characteristics related to economic deprivation, supporting their use in the proposed framework

With a focus on urban environments and housing inequality, Li et al. (2021) conducted research on urban poverty in China using high-resolution daytime satellite imagery. The study employed manual feature engineering to derive geometric, shape, and complex texture metrics from the imagery to identify different types of buildings and structural densities. These manually extracted spatial features were then fed into four traditional machine learning regressors: Random Forest, Gaussian Process Regression, Support Vector Regression, and a standard Neural Network. The empirical results revealed that the Random Forest model yielded the highest accuracy, achieving an R^2 of approximately 0.53. While the study concluded that structural characteristics derived from the Earth's surface can serve as dependable indicators of household economic situations, its reliance on manual texture extraction algorithms highlights the limitations of traditional machine learning compared to modern deep learning. This directly supports the necessity of the proposed CNN-LSTM architecture, which eliminates the need for manual feature engineering by automatically learning optimal spatial hierarchies.

Following a similar approach, Okaidat et al. (2021) demonstrated that using CNNs on satellite images gives much better results for poverty prediction than traditional statistical models, especially in cities where the infrastructure is complex. Their results support the main idea that one of the strong points of CNNs is their ability to identify features in large geospatial datasets.

Addressing the financial scalability of national-level poverty mapping, Ayush et al. (2021) proposed an efficient framework utilizing high-resolution remote sensing images and deep learning. Because acquiring sub-meter satellite imagery for an entire country is cost-prohibitive, the researchers implemented a novel Reinforcement Learning (RL) approach. Their model dynamically analyzed free, low-resolution imagery to intelligently select only the most informative regions where costly high-resolution imagery should be acquired. Upon acquiring the targeted high-resolution images of Uganda, the system utilized object detection networks to extract discrete economic indicators, which were subsequently mapped to poverty indices. By doing so, their model exceeded previous performance benchmarks while utilizing 80% fewer high-resolution images. While this study brilliantly addresses the cost-efficiency of spatial mapping, it focuses strictly on optimizing spatial acquisition rather than tracking continuous temporal changes. This further justifies the deployment of the proposed CNN-LSTM model, which relies on readily accessible data to extract both spatial features and historical, time-series trends.

Tang et al. (2024) developed a poverty estimation framework using a convolutional Long Short-Term memory (ConvLSTM) architecture that integrates multisource remote sensing data to estimate the wealth index (WI) in Nigeria at a 10×10 km spatial resolution. The study combined nighttime light (NTL) imagery, normalized difference vegetation index (NDVI), surface reflectance, land cover classification, and terrain slope data to capture environmental and socioeconomic indicators associated with poverty. Spatial features were first extracted using convolutional layers, while temporal dependencies from multiyear satellite observations were modeled through ConvLSTM to capture time-series patterns. The model's performance was evaluated using the coefficient of determination (R^2), where the ConvLSTM feature fusion model achieved an R^2 of 0.73, outperforming traditional approaches such as a cubic polynomial regression model using NTL data (R^2 = 0.36) and a standard CNN using NTL imagery (R^2 = 0.53). When only NTL time-series data were used, the ConvLSTM still improved prediction accuracy (R^2 = 0.66) compared with static CNN models. Despite its improved predictive capability, the study has limitations, including reliance on coarse 10 km grid resolution, potential spatial bias from survey-derived wealth index data, and dependence on remote sensing proxies that may not fully capture socioeconomic dynamics. These limitations suggest opportunities for future research using architectures that can further refine spatial–temporal feature learning and improve poverty estimation accuracy. (Tang et al., 2024)

2.6 Synthesis of Related Studies

The literature and studies reviewed in this chapter underscore the critical need for advanced, remote sensing-based poverty estimation systems, particularly in developing nations like the Philippines, where survey data is often infrequent or scarce. The findings reveal several key insights that strengthen the foundation of this study. The studies examined in the local literature highlight that in many cases, traditional survey methods fail to capture the most detailed and accurate image of poverty. The research carried out by the Asian Development Bank (2021) and Jimenez (2025) shows that physical features of the environment, such as the pattern of urbanization, human-made structures, and the layout of isolated homes, serve as meaningful indicators of poverty. Moreover, Aragoza and Reyes (2022) and Tingzon et al. (2019) demonstrated that nontraditional datasets such as nightlight intensity and internet speeds significantly complement spatial data in illustrating the economic activities of a specific area. Local research deeply aligns with the concept of this paper, thereby firmly establishing that satellite imagery and geospatial data are crucial instruments for policymakers in accurately identifying the poor in the Philippines.

Deep learning techniques have been proven highly effective for automating the analysis of highly detailed satellite images, as shown by several foreign studies. Various studies have revealed the exceptionally high accuracy in the prediction of Convolutional Neural Networks (CNNs) when used for recognizing socioeconomic indicators. Results from studies by Saeed and Turkoglu (2023), Li et al. (2021), and Okaidat et al. (2021) show that CNNs can detect complicated physical features, like building density, road networks, and types of roofing materials, that are strongly associated with household wealth. Additionally, Ayush et al. (2021) highlighted the potential of such models to be scaled up; CNN-based methods therefore allow for poverty mapping on a large scale, for instance, at the level of a country without any decrease in high-level prediction accuracy, which solves the problem of data collection by manual methods.

The appearance of hybrid deep learning models brings a whole new level to the forecasting of poverty by simultaneously focusing on spatial and temporal aspects. Srishti Gulecha et al. (2024) points out that CNNs are mainly used for capturing the spatial features, while LSTM models play an important role in identifying the temporal changes of socioeconomic factors. More specifically, their work shows that integrating different types of data such as spatial images and temporal sequences leads to a much better prediction capability than using one type of data only. Thus, the proposed framework not only locates the poverty areas but also has the potential to capture the dynamics of poverty over time.

The synthesis of the reviewed literature and studies reveals that a hybrid deep learning framework for poverty mapping in the Philippines is both viable and essential. Combining CNNs for spatial feature extraction and LSTM networks for temporal analysis offers an innovative and scalable method to overcome the limitations of traditional surveys. Applying advanced machine learning techniques, the present research seeks to close the gap between raw satellite imagery and usable socioeconomic data, thereby providing policymakers with a more dynamic tool for poverty-related decision-making.

Related Study	Algorithms Used for Feature Identification	Algorithms Used for Regression	Multi-source data (Daytime & Nighttime Satellite Images)	Temporal / Time-Series Analysis	Location
Asian Development Bank (2021)	CNN	Ridge Regression and Random Forest	Yes	No	Philippines
Tingzon et al. (2019)	CNN	Ridge Regression	Yes	No	Philippines
Ledesma et al. (2020)	N/A	XGBoost & Random Forest	No	No	Philippines
Jimenez (2025)	N/A	Generalized Least Squares, Support Vector Regression, Neural Network and Random Forest	Yes	No	Philippines
Srishti Gulecha et al. (2024)	CNN	Decision Tree Regressor, Random Forest Regressor, MLP	Yes	No	India
Tang et al. (2024)	CNN	LSTM	Yes	Yes	Nigeria
Proposed System (Project TALA)	CNN	LSTM	Yes	Yes	Philippines (trained nationwide; outputs for select localities)

Table 2: Summary and comparison of key related studies

Chapter 3

METHODOLOGY

3.1 Research Design

The researchers shall employ a quantitative and experimental research design. While the primary objective is cross-sectional – mapping poverty levels within a specific area– the research integrates a longitudinal data stream (multi-temporal satellite imagery and DHS survey data). This longitudinal approach, processed via a CNN-LSTM architecture, is critical for accounting for seasonal and atmospheric variations, such as cloud cover interference. By adopting an experimental research design, the researchers account for the characteristics mentioned, enabling a controlled environment to develop and assess the CNN-LSTM architecture and iteratively enhance the overall framework produced. Moreover, the quantitative approach in the research design allows systematical collection of data for both performance optimization and forecasting error metrics (e.g., R^2 and MSE). The quantitative approach provides stringency of statistical validation and ensures that results are both sound and generalizable.

3.2 System Architecture

The architecture follows a three-phase approach: (1) Proxy Task Pre-training, where a CNN is trained to predict nighttime luminosity classes (Dark, Dim, Bright) from daytime Sentinel-2 imagery; (2) Spatiotemporal Feature Extraction, where the weights from the pre-trained CNN are frozen and used to extract 4,096-dimensional feature vectors from quarterly satellite composites; and (3) Hybrid Regression, where an LSTM aggregates these temporal vectors and combines them with static OSM infrastructure features to estimate the DHS Wealth Index. The resulting outputs are then forwarded to the visualization module, which presents the results through an interactive dashboard user interface. Figure 2 shows the architecture of the system.

3.2.1 Activity Diagram

The activity diagram shown in figure 3 illustrates the complete workflow of the system, organized in three main phases: Data Acquisition, Modeling and Deployment. The process begins with the retrieval of the demographic health survey data, Sentinel 2 imagery, and VIIRS data. These datasets undergo coordinate alignment and clustering to ensure spatial consistency. Following this, the system extracts features from both daytime and nighttime data to train and then validate the hybrid CNN-LSTM architecture. This stage concludes the generation of the poverty mapping output. The final deployment phase outlines the integration of the trained model into the web-based visualization module. It details the user journey from website access to searching the interactive heatmap and finally downloading analytical results.

3.2.2 Use Case Diagram

The use case diagram defines the functional requirements, system scope, and interactions among the user, the CNN–LSTM model, and the visualization module. It outlines core system functionalities, including view mapping, which extends

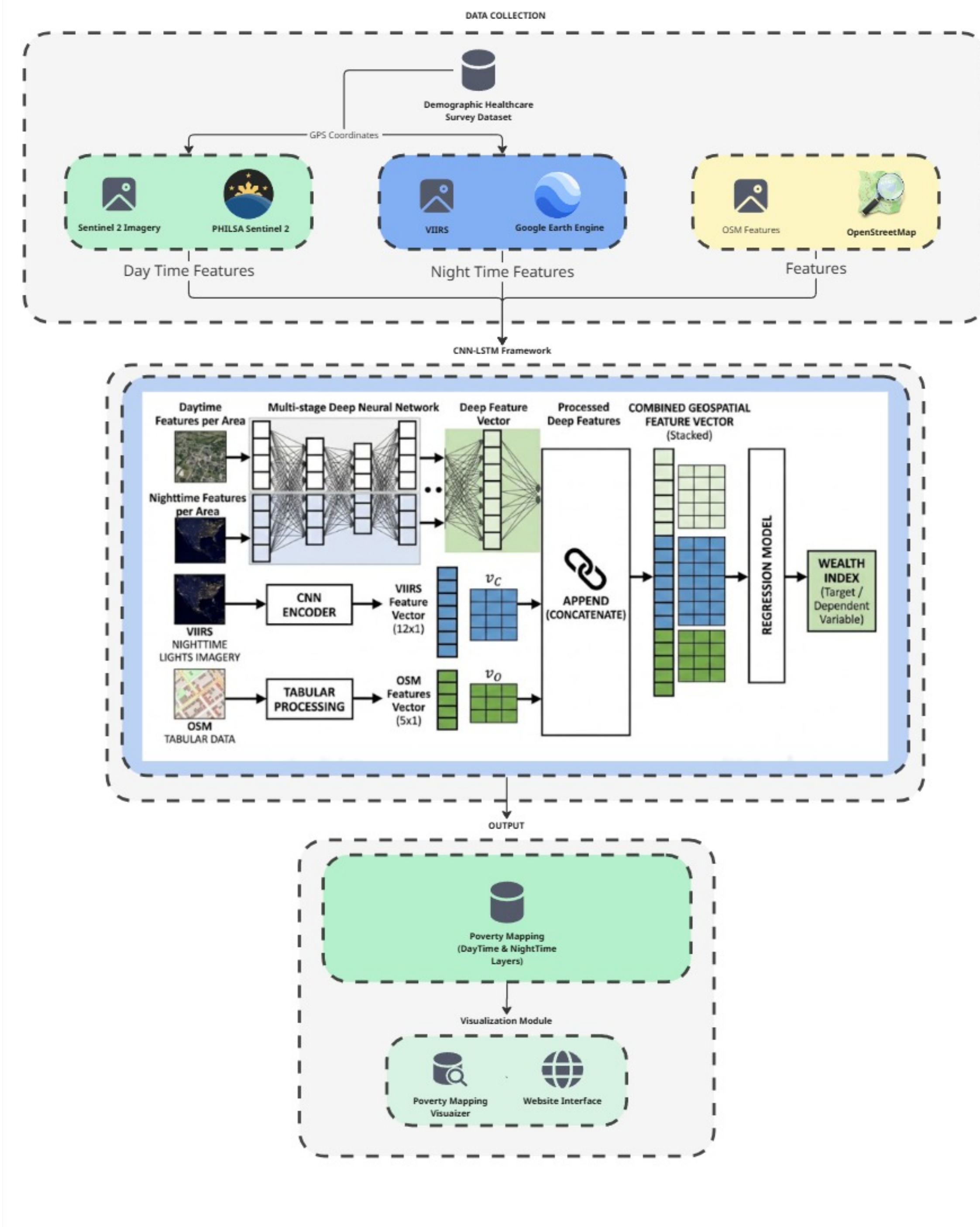

Figure 2: System architecture

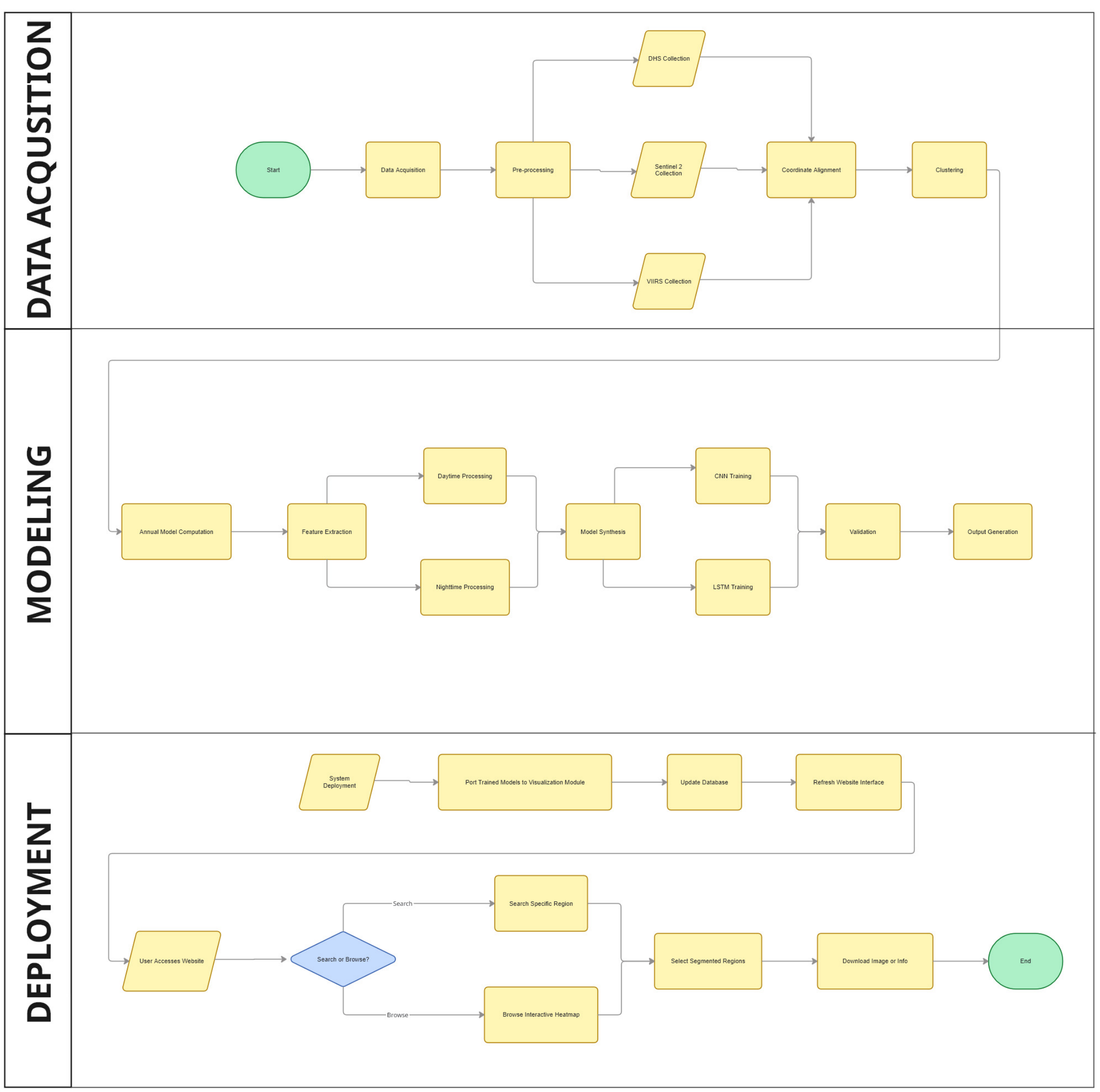

Figure 3: Activity diagram

search location capabilities; customization settings, which extend data uploading features; and user registration, which extends system guidelines such as authentication and session management.

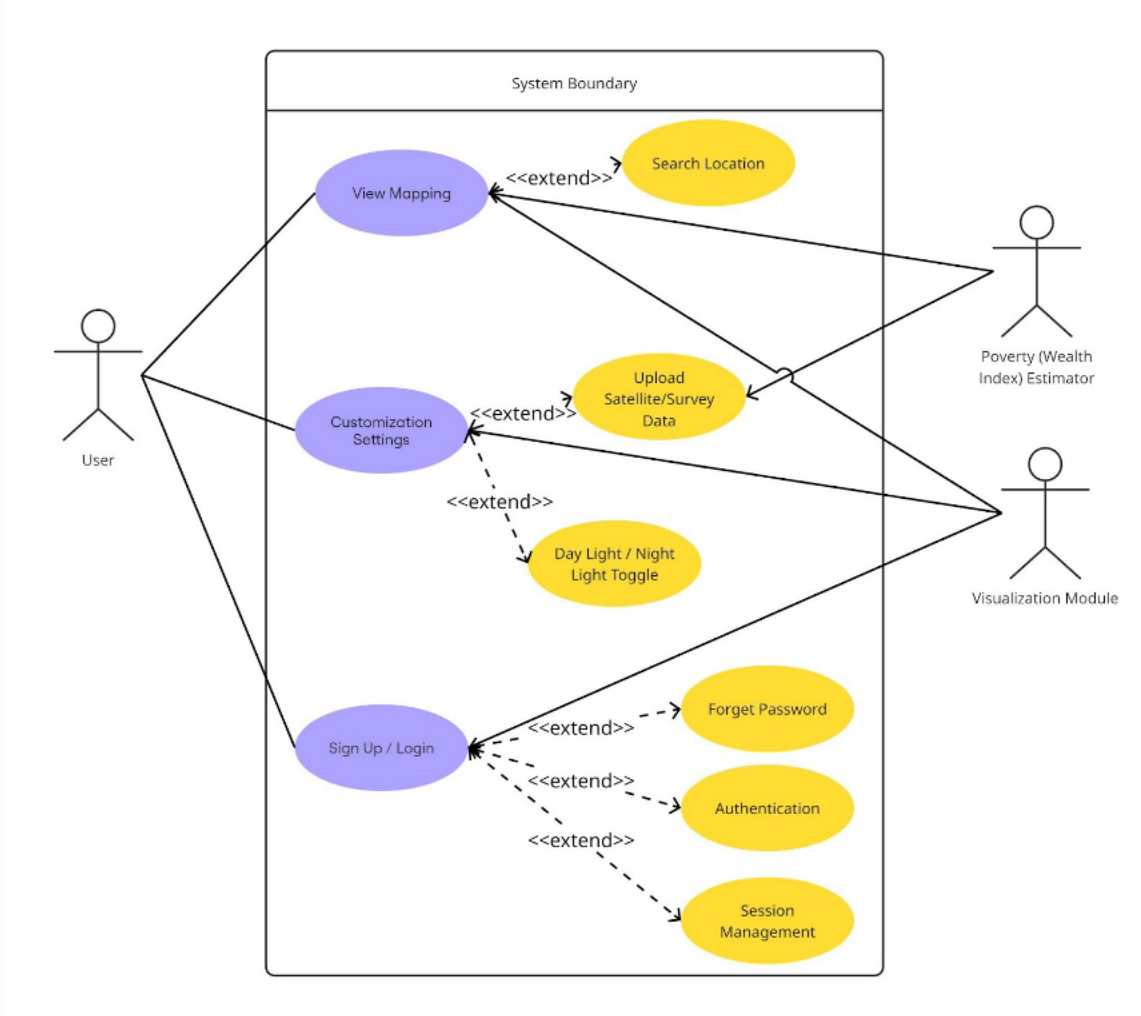

Figure 4: Use case diagram

3.2.3 Component Diagram

The component diagram visually represents the major system components and their interactions within the overall architecture. These components include data preprocessing, feature identification, regression modeling, and result generation and visualization.

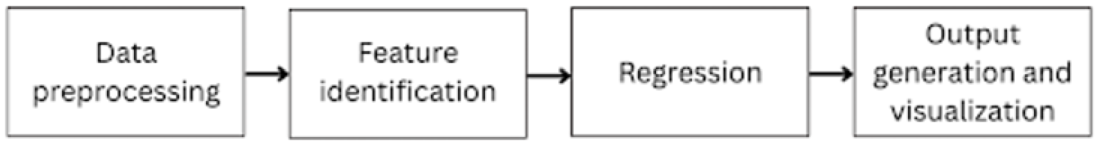

Figure 5: Component diagram

3.2.4 Deployment Diagram

The deployment diagram illustrates the procedural flow of the system across the deployment environment, highlighting how system components are distributed and interact during execution.

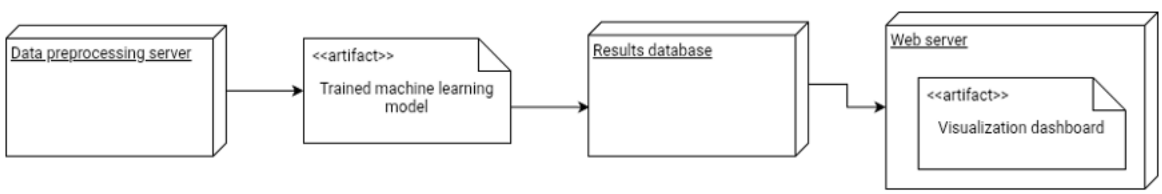

Figure 6: Deployment diagram

3.3 Hardware and Software Specification

The study is developed using Python 3.11.0 for its robust libraries for machine learning development and programming readability. Specifically, the study employs the following libraries to develop the CNN-LSTM framework: geemap, numpy, pandas, PIL, TensorFlow 2.15, and rasterio. To develop the visualization module the researchers have selected Next.js for its customizability and ease of integration.

The following specifications are the hardware used by the study:

Hardware Part	Model
CPU	Apple M4 (4 performance cores, 4 efficiency cores)
RAM	128 GB Unified Memory
GPU	Apple M4 10-core GPU (Unified Memory Architecture)
Storage	Apple MacBook Air Internal SSD

Table 3: Hardware specifications

The following specifications are the software environment used by the study:

Software Environment	Version
Operating System	macOS Tahoe (26.x)
ML Acceleration	Apple Metal Performance Shaders (MPS) via tensorflow-metal
Cloud Storage	Google Drive (2 TB)

Table 4: Software specifications

3.3.1 Computational Environment

The implementation pipeline operates across three computational environments, each serving a distinct role.

3.3.1.1 Google Earth Engine

Satellite image acquisition and preprocessing were conducted within the Google Earth Engine (GEE) cloud computing platform, accessed via the GEE Python API. GEE was selected because it provides server-side access to the complete Sentinel-2 and VIIRS image archives without local data storage or compute infrastructure, offloading image filtering, cloud masking, spatial compositing, and geographic clipping to Google's distributed processing environment. Exported GeoTIFF files were directed to a designated Google

Drive folder and to a Google Cloud Storage (GCS) bucket, which served as the primary data repository for downstream stages.

3.3.1.2 Google Colaboratory and Local GPU

TFRecord format conversion was performed in Google Colaboratory using a CPU runtime, as this is an I/O-bound sequential read operation that benefits from parallel file access rather than GPU computation. Proxy CNN training was conducted on a GPU runtime equipped with an NVIDIA T4 GPU (15 GB VRAM) using TensorFlow 2.19.0 with the MirroredStrategy distribution API. The final hybrid LSTM model training and hyperparameter search were conducted locally on an Apple MacBook Air M4 (128 GB unified memory) using the TensorFlow Metal backend for Apple Silicon GPU acceleration.

3.3.1.3 Google Cloud Storage

A dedicated GCS bucket in us-central1 served as the centralized data repository. GCS was selected for TFRecord storage because the TensorFlow tf.data API supports direct parallel reads from GCS without local mounting, essential for high-throughput GPU training pipelines. A secondary bucket in us-central2 was created for TFRecord replication to support Cloud TPU v4 workloads.

3.4 Dataset Collection, Preparation, and Features

To facilitate the development of the spatiotemporal deep learning architecture, this research uses various data from remote sensing, open-source mapping, and national surveys. This section details the distinct features and characteristics of the data sources utilized for model training and validation. All satellite imagery is obtained and preprocessed in close collaboration with the Philippine Space Agency (PhilSA).

3.4.1 Daytime Satellite Imagery: Sentinel-2

Daytime optical imagery was sourced from the Sentinel-2 MulitSpectral Instrument (MSI) Level-2A Surface Reflectance product, providing Bottom-of-Atmosphere (BOA) reflectance corrected for aerosol and water vapor influences. Five spectral bands were selected: B4 (Red), B3 (Green), B2 (Blue) at 10-meter native resolution; B8 (Near-Infrared) at 10 meters; and B11 (Shortwave Infrared) at 20 meters resampled to 10 meters during export. These bands were selected to capture visible surface characteristics, vegetation discrimination and built-up surface properties relevant to socioeconomic indicator estimation.

Images were organized into quarterly median composites spanning the calendar year: Q1 (January 1 to March 31), Q2 (April 1 to June 30), Q3 (July 1 to September 30), and Q4 (October 1 to December 31). Cloud masking was applied using the Sentinel-2 QA60 quality band encoding opaque cloud and cirrus cloud flags at bits 10 and 11 respectively; images with cloud pixel percentage exceeding 80% were excluded prior to compositing. For clusters where the cloud-filtered collection contained no valid observations in a quarter, a fallback mosaic from the unfiltered collection was generated. All images were exported at 10-meter spatial resolution using EPSG:3857 (Web Mercator Projection) for metric consistency across the Philippines' geographic extent from approximately 5°N to 21°N. The buffered spatial extent per cluster follows the adaptive DHS protocol: 2,000 meters for urban clusters and 5,000 meters for rural clusters. The buffered geometry was converted to a rectangular bounding box prior to export. A total of 4,988 GEE export tasks (1,247 clusters × 4 quarters) were submitted and confirmed completed.

3.4.2 Nighttime Light Imagery: Visible Infrared Imaging Radiometer Suite (VIIRS) DNB

Nighttime luminosity data was obtained from the VIIRS VNP46A1 daily gridded Day/Night Band product (NOAA/VIIRS/001/VNP46A1), providing at-sensor radiance in nanowatts per square centimeter per steradian (nW/cm²/sr) at approximately 500-meter resolution. A temporal median composite was computed across all daily 2022 observations per cluster, with a median reducer preferred over a mean to suppress transient light sources such as fishing vessel lights, fire events, and atmospheric airglow. Results were exported as a CSV via GEE's reduceRegions() function. The continuous annual median radiance values were discretized into three ordinal classes using pandas tertile boundaries: Class 0 (Dark, lowest tertile), Class 1 (Dim, middle tertile), and Class 2 (Bright, highest tertile), yielding an approximately balanced class distribution of 33.4% / 33.3% / 33.4% across 1,247 clusters.

3.4.3 OpenStreetMap (OSM) Infrastructure Data

Crowdsourced vector data from OpenStreetMap (OSM), accessed via the Geofabrik repository for the Philippines, provides explicit spatial labels to contextualize the daytime satellite imagery. Four OSM shapefile layers were used: roads, building footprints, points of interest, and landuse polygons. All layers were reprojected from EPSG:4326 to UTM Zone 51N (EPSG:32651) for meter-based distance and area calculations. Spatial indexing via the Rtree-based sindex.intersection() method was applied to each layer prior to the extraction loop, substantially reducing per-cluster query time.

3.4.4 Ground Truth: 2022 Philippines DHS Wealth Index

The ground-truth socioeconomic labels for the study are derived from the 2022 Philippines Demographic and Health Survey (DHS-VIII), conducted by the Philippine Statistics Authority (PSA) in collaboration with the United States Agency for International Development (USAID) DHS Program. The primary target variable is the Wealth Index, a continuous composite measure of a household's cumulative living standard. This index is constructed using Principal Component Analysis (PCA) based on variables found in the Household Recode (HR) file, including ownership of assets (e.g., televisions, vehicles, smartphones), source of drinking water, type of toilet facility, and flooring materials. The household-level Wealth Index Factor Score (variable hv271, scaled by 100,000) was divided by 100,000 and averaged across all households within each DHS cluster (hv001). This aggregation yielded 1,247 cluster-level values ranging from -2.369 to +1.874, with a national mean of +0.043 and standard deviation of 0.688. Spatially, the data is anchored to the Geographic Data (GE) file, which provides the GPS coordinates for each survey cluster. It is noted that these coordinates contain a standardized random displacement (jitter) of up to 2 kilometers in urban areas and 5 kilometers in rural areas to protect respondent anonymity, a factor that is methodologically accounted for through the use of cluster-buffer sampling.

3.5 Methods in Development

This study utilizes the Agile Scrum development technique that offers an iterative and incremental approach to software development. Agile Scrum development enhances flexibility, adaptability and continuous feedback from stakeholders. It is highly appropriate for this study as it provides quick prototyping and continuous improvement or changes to the system. This method gradually improves both the web-based visualization dashboard and the CNN–LSTM deep learning models to meet the changing analytical needs of policymakers.

The Agile development technique is comprised of several, short, time-bound cycles known as sprints that break down complex problems or tasks to smaller and simpler components. A sprint mainly comprises several steps such as ongoing development, testing, and evaluation. This iterative process contributes to enhanced software quality, higher prediction accuracy, and increased responsiveness to difficulties inherent in satellite data analysis, such as cloud interference and spatial resolution inconsistencies.

The key principles of Agile used in this study are:

1. Incremental development: The system is developed in functional components rather than as a whole. The spatial feature extractor (CNN) is implemented first, followed by the temporal analyzer (LSTM), and the final component is the visualization dashboard.

2. User-centric design: Potential end-users' (DSWD, DEPDev) and subject matter experts' feedback are incorporated into the interface of the dashboard to account for user's interpretability of the poverty maps for decision-making.

3. Continuous integration and testing: Each model component is validated iteratively with metrics such as R^2 and MSE to ensure high predictive accuracy before integration.

4. Flexibility and adaptability: The development process is responsive to changes in the model due to hardware performance and data availability.

5. Collaborative development: The team's coordination helps to ensure that data engineering, model training, and frontend development are working together.

This approach is consistent with the goals of the Poverty Incidence Estimator by allowing for standardized model training, hyperparameter optimization, and validation against ground-truth survey measure.

3.5.1 Scrum Methodology

The system is developed by following a full Agile Scrum workflow divided in 4 sprints. The process flow is described in the following steps:

1. Requirements Analysis and Planning

   Data Discovery: Which bands from Sentinel-2 (Optical) to use for VIIRS (Night Lights) to extract features? What OSM features to use? How to align the DHS survey clusters with the satellite pixels? How to handle cloud cover interference in the satellite imagery? What are the key performance metrics for model evaluation (e.g., R^2 for CNN, MSE for LSTM)? What are the user requirements for the visualization dashboard (e.g., heatmap interactivity, report generation)?

Hardware Configuration: Installing the development environment (Python, TensorFlow with tensorflow-metal) on the Apple M4 MacBook Air with 128 GB unified memory to ensure that the hardware meets the subsequent training and inference requirements.

User Stories: “As a policymaker, I want to toggle day/night layers to see economic activity”, “As a policymaker, I want to search for specific locations to view poverty estimates”, “As a policymaker, I want to download analytical reports for selected areas”.

2. Iterative Development Cycles (Sprints)

The project is divided in 4 sprints, each lasting two weeks. The development process in each cycle are as follows:

Sprint 1: Data Engineering and Preprocessing

(a) Activities: Aggregating quarterly median composites to remove cloud cover and aligning DHS survey clusters with satellite pixels. Concurrently, exporting full spatial coverage imagery from Google Earth Engine for select localities in the Philippines for inference and output generation.

(b) Deliverable: A cleaned, normalized multi-modal dataset ready for model training, alongside exported quarterly satellite composites for select localities ready for inference.

Sprint 2: Spatial Model (CNN) Development

(a) Tasks: Developing and training the Convolutional Neural Network on the proxy task of predicting nighttime luminosity classes from daytime Sentinel-2 imagery, capturing spatial indicators of poverty such as road density, building footprints, and land cover patterns. This includes hyperparameter tuning and validation of the proxy task performance.

(b) Deliverable: A trained CNN model with spatial features to be extracted

Sprint 3: Temporal Model (LSTM), Interpretability Pipeline, and Analytics Report Generation

(a) Tasks: Developing and training the LSTM network to learn from temporal quantities of poverty. Following model convergence, implementing the SHAP value computation pipeline using Kernel SHAP on the final regression layer. Developing the analytics report generation logic and formatting.

(b) Deliverable: A working hybrid CNN-LSTM inference engine with an integrated SHAP interpretability pipeline capable of producing analytics reports.

Sprint 4: Visualization Module

(a) Tasks: Developing the Next.js UI and integrating it with the poverty maps and the analytics report generation system. Implementing the report download functionality and the area-selection interface that allows users to request analytics reports for specific localities.

   (b) Deliverable: A fully working web application deployed on a web server for testing, featuring an interactive heatmap, an analytics report viewer, and a structured report download function.

3. Testing and Validation

   (a) Model Performance Testing: Evaluates the accuracy of the deep learning models using Coefficient of Determination (R^2) for the CNN and Mean Squared Error (MSE) for the LSTM.

   (b) Functionality Testing: Ensures that the Search Location and Layer Toggle features in the dashboard work as intended.

   (c) Usability Testing: Conducted with Subject Matter Experts (SMEs) to assess the system's ease of use using ISO 25010 standards.

   (d) Security & Compliance Testing: Ensures that the coordinate blurring (jittering) of DHS survey data is maintained to protect respondent privacy.

4. Deployment and Continuous Improvement

   (a) Deployment: Deploy the visualization module to a web server to allow remote access for evaluation.

   (b) Performance Monitoring: Continuously monitor the dashboard's load times and the model's inference speed on the deployment hardware.

   (c) Refinement: Update the hyperparameters or visualization scales based on feedback from the SME evaluation.

3.5.1.1 Scrum Roles

The study uses standard Scrum roles for effective execution:

- Product Owner: Establishes study goals and validates that the Wealth Index predictions have significance for the study.
- Scrum Master: Controls schedule, conducts daily stand-ups, and clears technical blocking issues (e.g. API limits or hardware errors).
- Development Team: Implements end-to-end solutions, including data preprocessing, training of deep learning models (CNN-LSTM), and full-stack web development.

Through this Agile Scrum workflow, the poverty estimation system is iteratively developed and refined through continuous stakeholder feedback, ensuring that both model accuracy and system usability meet the requirements of evidence-based governance in compliance with ISO 25010 quality standards.

3.6 Algorithms Used

3.6.1 Convolutional Neural Network (CNN)

The Convolutional Neural Network (CNN) functions as the primary engine for spatial pattern recognition. The logic relies on the mathematical operation of convolution, where learnable kernels (filters) slide over the input satellite imagery to generate feature maps.

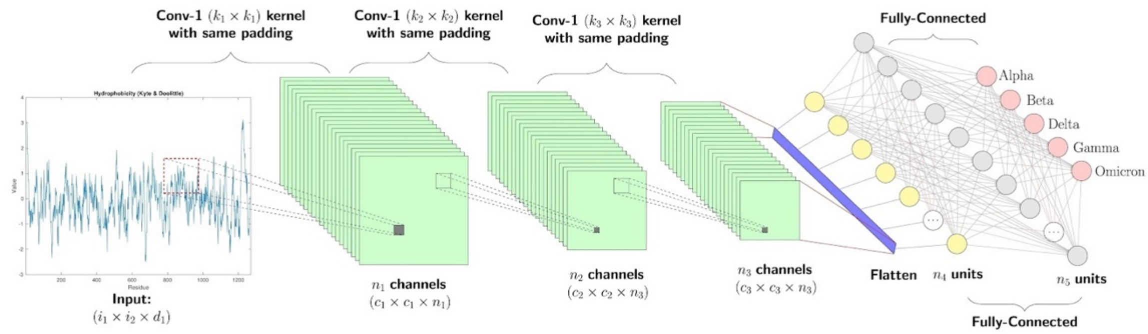

Figure 7: A convolutional neural network. Retrieved from https://www.authorea.com/users/918947/articles/1363525-development-of-a-machine-learning-model-for-automatic-detection-of-crop-diseases-case-study

The process follows a hierarchical sequence:

1. Convolution: Filters detect low-level features such as edges and textures in the early layers, and high-level semantic features such as road networks, building footprints, and urbanization patterns in deeper layers.

2. Activation: A non-linear activation function (ReLU) is applied to introduce non-linearity, allowing the network to learn complex data boundaries.

3. Pooling: Max-pooling layers downsample the spatial dimensions of the feature maps, reducing computational complexity while preserving the most invariant and salient spatial features.

4. Classification/Feature Output: For nighttime data, the network concludes with a classification head to categorize luminosity intensity; for daytime data, it outputs a flattened feature vector.

To overcome the limited sample size of the DHS survey, this study employs a transfer learning strategy. Instead of training the CNN directly on the wealth index, the model is first pre-trained on a proxy task: predicting VIIRS nighttime light intensity from daytime Sentinel-2 images. This forces the network to learn rich spatial representations such as identifying concrete roofs, road networks, and agricultural zones that correlate with light emission.

Once the model achieves satisfactory accuracy on the proxy task, the final classification layer is removed. The model then functions as a feature extractor, converting raw quarterly satellite images into abstract 4,096-dimensional feature vectors that serve as the input for the LSTM.

To ensure robust feature extraction, the CNN is optimized using Batch Normalization to stabilize learning and Dropout layers to prevent overfitting, particularly given the high dimensionality of satellite data. Validation is conducted by monitoring the categorical cross-entropy loss for the nighttime classification task and the feature variance for the optical data. The filters are updated via backpropagation using the Adam optimizer to minimize the error between predicted luminosity class and the ground-truth labels derived from VIIRS nighttime light intensity, enabling the model to learn spatial features without relying on sparse survey data.

3.6.2 Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) network is a specialized Recurrent Neural Network (RNN) designed to resolve the vanishing gradient problem in sequence modeling. Its logic centers on a cell state (long-term memory) regulated by three distinct gating mechanisms:

1. Forget Gate: Decides what information from the previous state is irrelevant (e.g., transient cloud cover or temporary seasonal vegetation) and should be discarded.

2. Input Gate: Determines which new information from the current time step (the dimensional feature vector extracted by the pre-trained CNN) is significant and should be stored in the cell state.

3. Output Gate: Computes the hidden state based on the filtered cell state, producing the final prediction vector.

The LSTM is integrated as the Temporal Aggregator in a “Many-to-One” configuration. It sits downstream from the CNN, ingesting the sequence of four concatenated feature vectors representing quarterly images. Its specific role is to analyze the temporal dynamics—specifically the stability of night lights and the consistency of built-up land cover—throughout the year. The final hidden state of the LSTM, which encapsulates the summarized annual socioeconomic context, is not used in isolation. Instead, it is concatenated with a static feature vector derived from OpenStreetMap such as road density, building counts, POIs and the annual VIIRS median. This combined vector is then fed into a final dense regression layer to predict the static DHS Wealth Index.

The LSTM is optimized using Backpropagation Through Time (BPTT), which unfolds the network across the four time steps to calculate gradients. Gradient clipping is employed to prevent exploding gradients, ensuring stable convergence. Validation of the hybrid CNN-LSTM-OSM architecture focuses on minimizing the Mean Squared Error (MSE) of the final regression output. The model's temporal generalization is validated using K-Fold Cross-Validation, ensuring that the learned spatiotemporal and infrastructural dependencies are consistent across different geographic regions of the Philippines.

3.6.3 SHAP (SHapley Additive exPlanations) Values

To ensure the transparency, interpretability, and operational utility of the hybrid CNN-LSTM model, this study incorporates SHAP (SHapley Additive exPlanations) values as an analytical method for generating per-area analytics reports. SHAP is grounded in cooperative game theory, specifically the Shapley values introduced by Shapley (1953), which fairly distribute a prediction's output among its contributing input features by computing each feature's average marginal contribution across all possible feature coalitions. Lundberg and Lee (2017) formalized this into a unified framework for machine learning interpretability, demonstrating that SHAP values satisfy three desirable properties: local accuracy, consistency, and missingness.

In the context of this study, SHAP values are applied to the final regression layer of the trained CNN-LSTM-OSM architecture. Since the model ingests a concatenated feature vector comprising CNN-extracted spatial embeddings, VIIRS nighttime luminosity statistics, and OSM infrastructure indices, SHAP values decompose each prediction of the DHS Wealth Index into attributable contributions from these input components.

Each analytics report contains the following components: a ranked feature importance table listing each input variable and its mean absolute SHAP contribution for the area, SHAP summary plots visualizing the distribution and direction of feature effects, SHAP dependence plots illustrating the relationship between key feature values and their attribution scores, and a narrative interpretation section generated to communicate findings in plain language accessible to non-technical stakeholders such as policymakers and community leaders. The resulting reports and visualizations are integrated into the visualization module's dashboard and are made available for download in a structured format to support offline policy analysis and reporting.

3.7 Method of Evaluating the Study

3.7.1 ISO 25010 Software Quality Evaluation

Figure 8: Software product quality attributes based on ISO 25010. Retrieved from https://iso25000.com/index.php/en/iso-25000-standards/iso-25010

Since the development of an effective system is paramount to the success of this study, the proponents are committed to ensuring software quality in strict adherence to ISO 25010 standards. The system evaluation is grounded in nine core attributes designed to verify its robustness and utility: (1) Functional Stability, (2) Performance

Efficiency, (3) Compatibility, (4) Interaction Capability, (5) Reliability, (6) Security, (7) Maintainability, (8) Flexibility, and (9) Safety. A detailed amplification of the specific characteristics associated with each of these attributes is illustrated in figure 8.

For this study, three critical attributes—Functional Suitability, Performance Efficiency, and Interaction Capability—are prioritized to ensure the system's effectiveness in estimating and mapping poverty levels using satellite imagery.

3.7.2 User Acceptance Evaluation

User Acceptance Testing (UAT) is an important stage of the software development life cycle, as it is the last step taken before the system deployment, and it is the stage when predefined test cases are taken to represent the way users are frequently performing their tasks and workflows in the system. During this stage, the system shall be tested on its end-users such as economists, social workers, and respective policymakers within bodies such as the Department of Social Welfare and Development (DSWD), Department of Economy, Planning, and Development (DEPDev), and other non-government organizations (NGOs) to assure them of the system meeting their needs of evidence-based governance and working under the real-world policy analysis environments.

3.8 Testing Method

3.8.1 Alpha Testing

The testing phase will be conducted in two steps, which are Alpha and Beta testing. Alpha testing will include in-house testing that the development team will use throughout the Scrum sprints to confirm the CNN feature extractor and LSTM temporal aggregator integration.

3.8.2 White Box Testing Procedure

The researchers will utilize white box testing in order to assess internal organization and logic of the system algorithms. This would include checking the data flow of the hybrid deep learning architecture that the quarterly satellite composites are properly passed through the Convolutional Neural Network (CNN) to extract features and then processed by the Long Short-Term Memory (LSTM) network to analyze the data in the temporal manner without internal errors.

3.8.3 Black Box Testing Procedure

The black box testing will help to examine how the system is functioning and not to examine the internal code but only the inputs and the expected outputs. The researchers will verify the adherence of the application to functional requirements, e.g., the possibility to search certain places, switch daytime and nighttime satellite images, and add new data, so that legitimate inputs give the right visual indicators of poverty on the dashboard.

3.9 Statistical Treatment of Data

The quantitative data collected in this study will be subjected to rigorous statistical analysis to ensure the validity and reliability of the findings. The analysis is divided into two distinct phases: (1) Predictive Model Evaluation, which assesses the accuracy of the satellite imagery-based poverty estimation, and (2) User Acceptance Evaluation, which assesses the usability and functionality of the developed web-based visualization module.

All statistical computations will be performed using Python, specifically the Scikit-learn and Tensorflow libraries for model evaluation and Microsoft Excel for user survey data analysis.

3.9.1 Predictive Model Evaluation

To determine the efficacy of the machine learning model in correlating satellite features such as nightlight intensity and spatial features with ground-truth socioeconomic data, the following metrics will be employed:

A. Coefficient of Determination (R^2)

The Coefficient of Determination, denoted as R^2, will serve as the primary metric to evaluate the regression model's utility. As defined by Draper and Smith (1998), R^2 measures the proportion of the total variation in the dependent variable, in this study, the wealth index, that is explained by the regression equation utilizing the independent variables which are satellite features. This statistic provides a clear summary of predictive capability; an R^2 value of 1.0 indicates that the fitted model explains all variability in the observed data, whereas a value of 0.0 implies that the model explains none of the variation (Draper & Smith, 1998).

$$R^2 = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2} \qquad \text{(Coefficient of determination)}$$

where $\hat{y}_i$ is the predicted value, y_i is the actual observed value, and $\bar{y}$ is the mean of the observed data.

B. Root Mean Square Error (RMSE)

Root mean square error will be used to measure the average magnitude of the error between the model's predicted poverty indices and the actual ground-truth values. A lower RMSE value indicates better model performance and higher accuracy. Unlike the Mean Absolute Error (MAE), RMSE squares the errors before averaging, which assigns a higher weight to larger errors. According to Chai and Draxler (2014), RMSE is the optimal metric for model evaluation when the error distribution is expected to be Gaussian.

Furthermore, Chai and Draxler (2014) argue that RMSE is preferable over Mean Absolute Error (MAE) in scenarios where large errors are particularly undesirable. Since RMSE squares the individual errors before averaging, it assigns a higher penalty to significant outliers. In the context of poverty estimation, this is critical: a large prediction error such as classifying a highly impoverished zone as wealth carries higher risks than minor deviations, making RMSE the more appropriate metric for ensuring model robustness.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \qquad \text{(Root mean square error)}$$

C. Pearson Correlation Coefficient (*r*)

To quantify the strength and direction of the association between the satellite-derived features and the ground-truth economic data, the Pearson Correlation Coefficient (r) will be calculated. As defined by Sedgwick (2012), this statistic measures the linear relationship between two continuous variables, providing a value between -1 and +1. A value of +1 indicates a perfect positive linear correlation, -1 indicates a perfect negative correlation, and 0 indicates no linear relationship.

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \qquad \text{(Pearson's correlation coefficient)}$$

The strength of the correlation will be interpreted based on the guidelines proposed by Cohen (2013): $|r| = 0.10$ to 0.29: Small/weak correlation, $|r| = 0.30$ to 0.49: Medium/moderate correlation, $|r| = 0.50$ to 1.00: Large/strong correlation.

D. SHAP-Based Localized Feature Attribution Analysis

To complement the quantitative performance metrics described above and to operationalize the model's outputs for policy use, SHAP (SHapley Additive exPlanations) value analysis will be conducted to generate localized analytics reports providing feature-level explanations of the model's predictive behavior for each geographic area. While R^2 and RMSE measure the aggregate accuracy of the regression output and Pearson's r quantifies the linear association between predicted and observed values, these metrics offer no insight into which input variables drive predictions for specific localities. SHAP values address this interpretability gap by decomposing each estimated Wealth Index value into additive contributions from each input feature.

The mean absolute SHAP value for each feature, denoted as $E[|\phi_i|]$, serves as the primary statistic within each report. A higher value indicates greater average influence on the model's poverty index predictions for that area, while a value approaching zero indicates negligible contribution.

3.9.2 Visualization Module User Acceptance and Evaluation

The assessment of the usability of the visualization module as perceived by stakeholders such as government agencies, local governments, non governmental organizations, and urban planners, will be analyzed using descriptive statistics.

A. Likert Scale

To quantify the qualitative perceptions of the respondents, the study will utilize the psychometric scale developed by Likert (1932). This 5-point scale is designed to measure the intensity of the respondents' agreement or disagreement with the usability statements derived from the ISO 25010.

B. Weighted Arithmetic Mean

To effectively summarize the central tendency of the Likert scale data, the

Scale	Range	Verbal Interpretation	Description
5	4.21 – 5.00	Strongly Agree	The feature is perfectly implemented and highly effective.
4	3.41 – 4.20	Agree	The feature is effective with minor room for improvement.
3	2.61 – 3.40	Neutral	The feature is acceptable but neither outstanding nor problematic.
2	1.81 – 2.60	Disagree	The feature is ineffective or difficult to use.
1	1.00 – 1.80	Strongly Disagree	The feature is completely non-functional or confusing.

Table 5: Likert Scale interpretation

Weighted Arithmetic Mean will be employed. As described by Cochran (1977), the weighted mean is a fundamental technique in survey sampling used to obtain an accurate estimate of the population mean when specific observations, or in this case, response categories, carry varying frequencies. By assigning weights corresponding to the frequency of respondents selecting a specific point on the Likert scale, this method ensures that the calculated average accurately reflects the collective consensus of the stakeholders regarding the system's usability.

$$\bar{x} = \frac{\sum fx}{n} \qquad \text{(Weighted arithmetic mean)}$$

where f is the frequency of each option chosen, x is the weight of each option (1-5), and n is the total number of respondents.

C. Standard Deviation

While the weighted mean indicates the average performance, the Standard

Deviation will be calculated to measure the variability or dispersion of the survey responses. Bland and Altman (1996) define the standard deviation as a crucial indicator of precision; in the context of this study, it serves as a measure of consensus.

$$SD = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}} \qquad \text{(Standard deviation)}$$

A calculated standard deviation value of less than 1.0 will be interpreted as a high level of agreement, indicating that the stakeholders' responses are clustered closely around the mean. Conversely, a standard deviation value greater than 1.0 will signify a divergence in opinion or a lack of consensus, characterized by widely scattered responses. Such dispersion suggests that the specific features in question may be polarizing and require further refinement to align with a broader range of user preferences.

D. Cronbach's Alpha

To validate the reliability and internal consistency of the survey instrument, Cronbach's Alpha will be computed. Originally introduced by Cronbach (1951), this coefficient estimates the reliability of a psychometric test by measuring how well a set of questions measures a single latent construct which in the case of this study is usability.

$$\alpha = \frac{K}{K-1}\left(1 - \frac{\sum \sigma_{y_i}^2}{\sigma_x^2}\right) \qquad \text{(Cronbach's alpha)}$$

A value of $\alpha \geq 0.70$ will be considered acceptable, confirming that the survey items demonstrate internal consistency and that the collected data is robust enough for subsequent statistical inference.

3.10 Respondents of the Study and Sampling Technique

The respondents of this study will be based on expertise, involvement in policymaking, and policy enforcement. A total of 25 respondents will be involved and categorized as follows: 10 technical experts, 10 policynakers and LGU representatives, and 5 economists. The said respondents will be asked to answer ISO25010 questionnaires to evaluate the system.

A purposive sampling technique shall be employed to ensure that the respondents will meet the level of specialized knowledge relevant to the study. This will allow the researchers to collect impactful feedback from individuals with domain expertise and evaluate the system's practicality, accuracy, and real-world applicability. Utilizing a distributed survey questionnaire, the respondents shall assess the effectiveness of the system in its predictive performance to visualize poverty clusters and estimate poverty levels.

Respondent Type	No. of Respondents
Technical Experts	10
Policymakers/LGU Representatives	10
Economists	5
Total Number of Respondents	**25**

Table 6: Distribution of respondents

Chapter 4

RESULTS AND DISCUSSION

4.1 Dataset Collection and Preprocessing

4.1.1 Daytime Satellite Image Processing

Prior to CNN training, all GeoTIFF images were converted to TFRecord, TensorFlow's binary serialization format, enabling the tf.data API to read training examples in parallel from multiple file shards. The 4,988 images were distributed across 32 TFRecord shards. Each TFRecord example stored serialized image bytes, the integer VIIRS class label (0, 1, or 2), the cluster ID, the quarter index, and image dimension metadata. During conversion, the first three spectral bands (B4, B3, B2) were stacked into a three-channel float32 array and normalized by percentile stretch: pixel values were rescaled to [0.0, 1.0] using 2nd and 98th percentile bounds, computed as normalized = $clip((img - p_2)/(p_{98} - p_2), 0, 1)$. This approach is robust to extreme pixel values from cloud edges and sensor artifacts. Images were resized to 224×224 pixels via bilinear interpolation. Images with persistent cloud cover producing near-constant pixel values $(p_{98} \leq p_2)$ were excluded via NaN guard condition.

4.1.2 Static Feature Engineering

A total of 52 static features were engineered from OSM and VIIRS data per prediction point buffer, organized into six thematic groups as detailed in Table xx.

4.1.2.1 Road Feature Extraction

Road geometries were classified by OSM fclass into four categories: main roads (motorway, trunk, primary, primary_link), secondary roads (secondary, tertiary), local roads (residential, living_street, unclassified, service), and tracks (track, path). Segments intersecting the buffer were clipped and cumulative length computed in km. Total_Road_Length aggregates all classified and unclassified segments. Extracted total road length per cluster ranged from 0.0 to 1,448.8 km, with a mean of 146.1 km, reflecting the wide variation in road network density between dense urban centers and remote rural clusters.

4.1.2.2 Building Feature Extraction

Building footprints were classified by OSM fclass into five types: residential, commercial, industrial, school, and hospital. For each type and in aggregate, four metrics were computed: total count, total footprint area (m^2), mean footprint area (m^2), and proportion of buffer area covered. Building counts ranged from 0 to 136,450 per cluster. High zero rates in typed building columns (approximately 57–60%) are consistent with known incompleteness of building type attribution in the Philippine OSM dataset and do not reflect processing errors.

4.1.2.3 Point-of-Interest and Landuse Feature Extraction

POIs were filtered to fifteen verified service category fclass values present in the Philippine OSM extract. Landuse polygons were classified into five categories (residential, commercial, industrial, agricultural, forest) and total area in m^2 within each buffer was summed. Landuse area features, along with POI_restaurant_Count and Total_POI_Count, were log1p-transformed before standardization to reduce right skew, with this transformation applied consistently at both training and inference time. All 52 features contained no missing values across 1,247 training clusters.

4.1.2.4 Feature Normalization and Aggregation Semantics

All 52 features were standardized using a fitted StandardScaler, with parameters estimated on the training set and applied without refitting at inference. For province and municipality-level reporting in the visualization module, sum features (road lengths, building counts, areas, POI counts, landuse areas) are reported as area totals aggregated across all prediction point buffers within the geographic unit. Mean features (Mean_Bldg_Area, building density, building type proportions, and VIIRS_Median) are reported as per-buffer averages.

4.2 CNN Proxy Task Pre-training

4.2.1 Architecture and Transfer Learning Strategy

The CNN serves as the spatial feature extractor for the hybrid architecture. A transfer learning strategy is employed: the network is first pre-trained on the proxy VIIRS luminosity classification task using the larger 4,988-tile dataset, before being used as a feature extractor for the 1,234-cluster wealth regression task. VGG16, originally pre-trained on ImageNet, serves as the convolutional base. The fully-connected classification layers are replaced with randomly initialized fully convolutional top layers following Xie et al. (2016), accommodating 224×224 pixel input tiles of arbitrary spatial extent. Only the last convolutional block is fine-tuned; all preceding layers are frozen.

4.2.2 Proxy Task Training Configuration

The proxy task is three-class VIIRS nighttime luminosity classification (Dark, Dim, Bright). Batch Normalization and Dropout are applied throughout the convolutional stack. Training uses categorical cross-entropy loss and the Adam optimizer with an initial learning rate of 10^{-6}, batch size of 32, and a maximum of 50 epochs. The learning rate is reduced by a factor of 10 on validation loss plateau. The proxy task achieves a final validation accuracy of 71.97%, confirming that the learned spatial representations meaningfully encode spatial features associated with economic activity.

4.2.3 Feature Extraction and Dimensionality Reduction

Upon completion of proxy task training, the classification head is removed and the CNN operates as a frozen feature extractor producing 4,096-dimensional activation vectors. Of 4,988 cluster-quarter records, 4,939 (representing 1,236 unique clusters) were successfully extracted; 49 missing records correspond to persistently cloud-obscured cluster-quarter pairs excluded by NaN guard. PCA was applied to reduce the 4,096-dimensional vectors to 256 principal components, fitted on the training feature array of shape (1,234×4, 4,096). The 256-component solution retains 90.87% of total explained variance, removing collinear and redundant CNN dimensions while substantially reducing LSTM parameter count and overfitting risk.

4.3 Hybrid CNN-LSTM Architecture

The final regression model is a dual-branch architecture processing temporal and static features in parallel before fusing them for regression. Table xx summarizes the complete configuration.

A. Temporal Branch: LSTM

The LSTM ingests four 256-dimensional PCA-compressed CNN feature vectors and outputs a 64-dimensional hidden state from the final time step, capturing the annual spatiotemporal summary of quarterly satellite observations. L2 regularization (1e-5) is applied to kernel weights; a Dropout layer (0.4) follows the output. The LSTM is optimized using Backpropagation Through Time (BPTT) with gradient clipping.

B. Static Branch: Dense Network

The 52 standardized static features are processed by a 64-unit dense layer (ReLU, Batch Normalization, Dropout 0.2, L2 1e-5). The 64-dimensional LSTM output and 64-dimensional static embedding are concatenated into a 128-dimensional representation, passed through a 64-unit fusion dense layer (ReLU, Dropout 0.2, L2 1e-5), then through a single-unit linear output layer producing the scalar wealth index prediction.

4.3.1 Model Training and Validation

4.3.1.1 Hyperparameter Optimization

Hyperparameters were selected via structured search evaluated on a held-out 20% validation split (987 training, 247 validation clusters) using best validation MSE as criterion. The best configuration: LSTM units = 64, dynamic dropout = 0.4, static dense units = 64, static dropout = 0.2, fusion units = 64, fusion dropout = 0.2, L2 regularization = 10^{-5}, learning rate = 10^{-3}. This configuration achieved a best validation MSE of 0.2304.

4.3.1.2 Cross-Validation Protocol

Five-fold cross-validation was applied to the full 1,234-cluster dataset. Within each fold, PCA was re-fitted on the training split and applied to validation to prevent data leakage; likewise for the StandardScaler. Each fold trained for up to 200 epochs with EarlyStopping on validation MSE loss (patience = 15) and ReduceLROnPlateau (factor = 0.5, patience = 7, minimum learning rate =

10^{-6}).

4.3.1.3 Final Model Persistence

A final model was trained on all 1,234 clusters using the best hyperparameter configuration, with a 10% internal validation split for EarlyStopping (patience = 20) and ModelCheckpoint restoring the best-epoch weights. The model, scaler, and PCA transformer were saved, constituting the complete inference artifact set for 2025 output province prediction.

4.3.2 Inference from 2025 Data

A spatially uniform prediction grid was generated across sixteen target areas: five provinces with the highest PSA 2023 poverty incidence as seen in table 1, the National Capital Region, and five additional localities selected to provide geographic and socioeconomic representativeness across Central Luzon and MIMAROPA: Bulacan, Bataan, Zambales, Oriental Mindoro, and Occidental Mindoro. Grid points were generated at 5,000-meter spacing in rural boundaries and 2,000-meter spacing within NCR municipalities. For DHS-origin clusters, the Urban_Rural classification is carried directly from the DHS cluster GPS displacement record (U for urban, R for rural), consistent with the DHS survey protocol. For grid-generated points, Urban_Rural classification is approximated from municipal boundary intersection. DHS cluster GPS locations within all sixteen study areas were added directly; grid points within 2,000 meters of any DHS location were removed to prevent spatial redundancy. The final prediction set comprises 2,000 points (1,723 grid-generated, 277 DHS-sourced) distributed across the sixteen study localities.

Province-scale Sentinel-2 imagery for 2025 was exported from GEE producing 64 GeoTIFF files (16 localities × 4 quarters, ranging 404 MB to 3,293 MB each). Each prediction point tile was extracted using rasterio's WarpedVRT interface for a 10 km × 10 km bounding box centered on the prediction point coordinates. A total of 1,970 clusters yielded complete four-quarter tiled imagery and were successfully processed through the full CNN feature extraction and LSTM inference pipeline, comprising 1,653 grid-generated points and 317 DHS-origin clusters. Points not processed are concentrated at provincial boundary edges where the bounding box exceeds the GeoTIFF export extent. Static OSM and VIIRS features were extracted for all prediction points using the same adaptive buffers, log1p transforms, and StandardScaler applied during training.

4.3.3 Results from 2025 Data

4.3.3.1 Coverage and Geographic Distribution

Of the 2,000 prediction points generated across the sixteen study localities, 1,970 clusters were processed through the full CNN-LSTM inference pipeline after incorporating DHS-origin clusters within each locality's boundary. The total of 1,970 comprises 1,653 grid-generated points and 317 DHS-origin survey clusters for which the actual DHS Wealth Index is available as a ground-truth reference. Points not processed are concentrated at provincial boundary edges where the 10 km × 10 km prediction tile exceeds the GeoTIFF export extent, a pattern considered satisfactory as missing points are systematically located at geographic margins.

The 1,970 inference clusters are distributed across the sixteen localities as follows, sorted from poorest to wealthiest by mean predicted wealth index: Basilan (42), Tawi-Tawi (29), Occidental Mindoro (242), Davao Oriental (194), Zamboanga del Norte (239), Maguindanao del Sur (83), Oriental Mindoro (157), Kalinga (110), Bulacan (95), Ilocos Norte (119), Aklan (61), Zambales (174), NCR (149), Pampanga (109), Benguet (112), and Bataan (55). The larger point counts in Occidental Mindoro, Zamboanga del Norte, Davao Oriental, and Zambales reflect their greater land area and the 5 km rural grid spacing applied, while the lower counts in Tawi-Tawi and Basilan reflect their smaller geographic extents and predominantly maritime territory.

4.3.3.2 Predicted Wealth Index Distribution

Predicted wealth index values across the 1,970 inference points range from -2.2389 to +0.693, with a mean of -0.585, a median of -0.546, and standard deviation of 0.543. The predominantly negative distribution, with 83.9% of points below zero, reflects the geographic composition of the study locality set: the five poorest provinces contribute 604 of the 1,970 inference points, biasing the aggregate distribution toward the lower end of the wealth index scale. The predicted range is compressed relative to the training target range of -2.369 to +1.874, reflecting both the deliberate selection of extreme-poverty and low-poverty provinces (which do not span the full national wealth range) and the well-documented regression-toward-mean behavior of regularized models at extreme values. The underestimation of high wealth values for NCR and the wealthiest provinces may additionally reflect the temporal distribution shift between 2022 training imagery and 2025 inference imagery.

4.3.3.3 Spatial Patterns of Predicted Wealth

Figure 4.2 presents a scatter plot of predicted WI against actual DHS survey WI for all 317 DHS-origin clusters processed through the full inference pipeline. The scatter yields a pooled R^2 of -0.5765 and Pearson r of 0.477, with points colored by province to reveal geographic bias in residuals. The negative pooled R^2 indicates that the model's absolute predictions are systematically offset from the 2022 survey values on this out-of-sample set, an expected consequence of the 2022-to-2025 temporal gap and the geographic novelty of several localities relative to the training distribution. Out-of-sample validation yields a mean absolute error of 0.775 and an RMSE of 0.946. These substantially exceed the cross-validation metrics (RMSE = 0.473, r = 0.727), as expected: five of the sixteen localities have no training clusters in the DHS dataset, the temporal gap introduces genuine wealth changes the model cannot observe, and DHS GPS displacement adds irreducible positional noise. Per-locality MAE ranges from 0.339 (Maguindanao del Sur) to 1.090 (NCR); the elevated NCR MAE is partially attributable to its concentration of DHS clusters (126 of 317, or 39.7% of all DHS-origin points) and the rapid pace of urban development between 2022 and 2025. The Pearson r of 0.477 confirms that the ordinal ranking of predicted wealth across clusters maintains a statistically significant positive correlation with ground-truth survey wealth (p ¡ 0.001), consistent with the poverty mapping literature noting that cross-geography transfer preserves relative ordering more reliably than absolute value accuracy. (Yeh et al., 2020; Jean et al., 2016).

At the provincial level, predicted wealth gradients broadly align with the known socioeconomic geography of the Philippines. Within NCR, the highest predicted wealth values are concentrated in central commercial districts, with lower predictions in coastal fringes and informal settlement zones. Within Zamboanga del Norte and Maguindanao del Sur, predicted wealth is lowest in upland and remote coastal municipalities with limited road access and sparse OSM-documented infrastructure. Within Ilocos Norte and Pampanga, higher predicted wealth is observed in urbanized municipal centers compared to agricultural hinterlands.

The spatial heterogeneity of predictions within individual provinces represents a primary analytical output of the study. Even within provinces designated as poor or wealthy at the aggregate level, substantial sub-provincial variation is captured, identifying specific municipalities with predicted wealth profiles that diverge significantly from the provincial mean. This municipality-level spatial granularity, produced without household surveys, represents the principal practical contribution of the study to poverty mapping in the Philippine context, enabling targeted resource allocation toward specific municipalities rather than blanket province-level interventions.

4.4 Model Performance

4.4.1 CNN Proxy Task Performance

The VGG16 CNN pre-trained on the VIIRS nighttime luminosity proxy classification task achieves a validation accuracy of 71.97% on the three-class Dark/Dim/Bright schema. This performance confirms that the learned convolutional representations meaningfully encode spatial patterns associated with nighttime light emission, including road network density, building coverage, and land cover composition that together determine a cluster area's luminosity signature. These representations provide the foundation for wealth index estimation in the downstream LSTM regression stage. The achieved accuracy is consistent with Tingzon et al. (2019), who reported 72% validation accuracy for a similar Philippine proxy task using VGG16, and with Xie et al. (2016), who reported 60% F1 on a five-class schema for sub-Saharan African clusters. The modest ceiling reflects inherent ambiguity in three-class luminosity thresholds across a geographically diverse landscape, but is sufficient to produce feature representations meaningfully correlated with the continuous DHS Wealth Index.

4.4.2 Hybrid CNN-LSTM Cross-Validation Performance

Table xx presents the five-fold cross-validation results of the final hybrid CNN-LSTM model trained on 1,234 DHS survey clusters with the expanded 52-feature static branch and 256-dimension PCA-reduced temporal features.

The model achieves a mean R^2 of 0.5244 ± 0.025, a mean RMSE of 0.4729 ± 0.018, and a mean Pearson r of 0.7271 ± 0.016. The pooled R^2 of 0.5269 and pooled r of 0.7259 are consistent with fold-level means, confirming stable generalization without outlier folds. The Pearson r of 0.7259 classifies as a large effect size under Cohen's (2013) guidelines ($|r| \geq 0.50$), indicating a strong positive linear relationship between predicted and observed DHS Wealth Index values across held-out clusters.

The low standard deviation across folds, particularly ±0.016 on Pearson r, confirms consistent generalization across different geographic partitions of the training data. This stability is attributable in part to the PCA dimensionality reduction, which removes noisy high-dimensional CNN components that caused erratic fold performance in preliminary experiments.

4.4.3 Contextual Benchmarking

The R^2 of 0.5269 is competitive with published results for single-country poverty mapping studies. Tingzon et al. (2019) reported an r^2 of 0.63 for the Philippines using CNN features, OSM, and nighttime light data in a ridge regression framework, though that study used multiple DHS survey waves yielding a substantially larger labeled dataset. (pls verify if true) Xie et al. (2016) reported R^2

ranging from 0.51 to 0.75 across five African countries using a comparable CNN transfer learning strategy. (pls verify if true. Add another study, preferably yung nigerian study na may cnn-lstm din) The present study's performance must be interpreted in the context of its constraints: a single 2022 DHS survey wave providing only 1,234 labeled clusters, geographic heterogeneity spanning densely urbanized NCR and remote island provinces simultaneously, and irreducible label noise from the DHS GPS jitter protocol.

4.4.4 Performance Constraints and Discussion

Three principal constraints limit the achievable cross-validation performance. First, training set size: deep learning models are sensitive to training set size relative to input feature dimensionality, and 1,234 clusters is small for a model with approximately 125,000 trainable parameters, even with the regularization measures applied. Studies reporting higher performance typically leverage pooled multi-country datasets totaling several thousand labeled clusters.

Second, geographic heterogeneity: the Philippine archipelago encompasses highly urbanized NCR districts, densely forested Cordillera provinces, and remote BARMM island provinces within a single training distribution. The model must generalize across fundamentally different land cover types, building typologies, and economic activity signatures, substantially increasing the regression task's difficulty compared to geographically homogeneous study areas. Folds containing higher proportions of BARMM clusters, where OSM coverage is sparser and VIIRS signals are consistently low, may exhibit reduced feature discriminability.

Third, DHS GPS displacement introduces irreducible label noise: cluster coordinates are displaced up to 2 km (urban) and 5 km (rural) to protect respondent anonymity, so imagery and OSM features extracted for each cluster may not correspond precisely to the residential area of the surveyed households. This geographic imprecision places an upper bound on achievable predictive accuracy from location-based features that is independent of model architecture.

4.5 SHAP-Based Interpretability Analysis

SHAP values are computed for all 1,970 prediction points using Kernel SHAP applied to the final regression layer of the trained model via the SHAP Python library. The explainer uses a background dataset of 200 randomly sampled training clusters and 500 coalition samples per instance, with l1_reg='aic' for sparse attribution regularization. The SHAP base value is approximately -0.58, the mean predicted Wealth Index across the 16 study localities. The 116-dimensional input (64 LSTM dimensions + 52 static features) decomposes into six interpretable categories: Temporal (LSTM dimensions), Satellite (VIIRS_Median), Road, Building, POI, and Landuse.

4.5.1 Temporal versus Static Feature Contribution

Across all 1,970 prediction points, LSTM contributions, representing the temporal satellite signal encoded from quarterly Sentinel-2 imagery, account for 53.1% of mean total absolute SHAP contribution per point. The 52 named static features collectively contribute the remaining 46.9%. This near-equal split distinguishes the final model from simpler single-branch architectures (mention other studies with single branch here) and reflects the expanded static feature set, which provides the static branch sufficient discriminating power to compete meaningfully with the temporal satellite signal. The inclusion of landuse polygon areas, fifteen POI categories, building type-specific metrics including per-type mean areas, and building density metrics substantially broadens the information available to the static branch relative to a road-and-building-count schema.

The temporal features nonetheless remain the individually largest driver of predictions on a per-feature basis: a single LSTM dimension encodes a compressed nonlinear summary of 4,096-dimensional CNN activations across four quarters, capturing land cover seasonality, vegetation phenology, built-up surface stability, and nighttime light dynamics simultaneously. The static features each capture a single observable dimension of infrastructure or landuse. The finding that LSTM contributions account for 53.1% of total absolute SHAP, despite operating on a 256-dimensional PCA-reduced representation, underscores the information density of the temporal satellite signal. The CNN-LSTM pipeline effectively encodes spatial patterns from quarterly Sentinel-2 composites into a temporal representation that carries more predictive information per feature than any available static geospatial indicator.

4.5.2 Top Static Feature Drivers

Table xx presents the top ten static feature drivers by mean absolute SHAP value across the 1,970 inference points.

4.5.3 Landuse as the Dominant Static Driver

The emergence of `LU_Agricultural_m²` as the highest-ranked static feature (mean $|SHAP| = 0.083$) reflects the geographic composition of the study provinces. The five poorest provinces in the study are predominantly rural and agricultural, while the wealthiest have lower agricultural landuse coverage and higher commercial and residential landuse. Agricultural landuse area therefore functions as a geographic separator between subsistence farming communities and wealthier urbanized or commercial-sector areas. This is consistent with the established relationship between land use type and asset wealth in the Philippines, where smallholder agricultural households systematically exhibit lower DHS Wealth Index scores. The inclusion of landuse polygon areas, absent from earlier road-and-building-count static feature schemas, is a primary factor in the elevated static branch SHAP contribution in the final model.

`LU_Forest_m²` ranks third (mean $|SHAP| = 0.053$), with 54.2% of non-zero SHAP values negative, reflecting that forest-dominated areas correspond to upland and island communities with limited road access, sparse built-up infrastructure, and low VIIRS luminosity. These patterns are consistent with poverty concentration in forest-adjacent municipalities in Kalinga, Benguet, and the BARMM island provinces.

`LU_Commercial_m²` ranks sixth (mean $|SHAP| = 0.036$) and exhibits a predominantly negative SHAP direction (69.1% of non-zero values negative). This directional finding warrants attention: commercial landuse in the Philippine OSM dataset commonly includes wet market areas and small commercial zones in lower-income municipalities, where zoned commercial land co-exists with limited built infrastructure and sparse POI density. The feature therefore functions as a marker of designated but underdeveloped commercial activity rather than a uniformly positive indicator of economic development. `LU_Residential_m²` (rank 9, mean $|SHAP|$ = 0.027) and `LU_Industrial_m²` (rank 8, 0.029) round out the landuse category, with industrial area contributing positively in export-processing zone clusters in Bataan and Zambales and negatively in informal industrial areas adjacent to poor communities.

4.5.4 Service Facility, Infrastructure, and Education Features

`Total_POI_Count` ranks second (mean $|SHAP|$ = 0.061), capturing aggregate economic footprint as a proxy for commercial activity and market access. High POI density is associated with formal economic integration and consumer purchasing power, both of which are strongly correlated with DHS Wealth Index scores.

The appearance of `Bldg_school_MeanArea` at rank 4 (mean $|SHAP|$ = 0.048) is a particularly informative finding enabled by the expanded building type-specific metrics. Mean school building footprint area serves as a proxy for education infrastructure quality: larger school structures indicate more formal, better-resourced educational establishments associated with areas of higher public investment and sustained economic development. This distinction between infrastructure quality and mere facility presence is only capturable through per-type MeanArea metrics introduced in the final 52-feature schema. Restaurant count (rank 10, mean $|SHAP|$ = 0.022) reflects consumer spending capacity and the presence of a discretionary dining economy.

`Bldg_residential_MeanArea` at rank 7 (mean $|SHAP|$ = 0.031) provides a complementary quality-of-housing signal: larger residential building footprints indicate more substantial dwelling structures associated with higher household asset accumulation. `VIIRS_Median`, despite its central role in the proxy CNN training task, ranks 30th among the 52 static features (mean $|SHAP|$ = 0.005), confirming that its luminosity signal is substantially subsumed by the LSTM temporal encoding of quarterly Sentinel-2 imagery and the expanded landuse feature set.

4.6 Visualization Module: Project TALA Web Application

The visualization module is a full-stack web application built with Next.js 14, TypeScript, Tailwind CSS, Mapbox GL JS (interactive map), Chart.js (analytics charts), html2canvas and jsPDF (PDF generation), and NextAuth.js (authentication). A Python data preparation script generates six JSON files consumed by the application: points.json (1,970 prediction points with wealth index, DHS WI where available, and per-point OSM values), provinces.json (16 provinces with aggregated statistics), municipalities.json, shap_global.json (global SHAP rankings), shap_by_area.json (per-province top-15 SHAP features), and national_stats.json (study-wide descriptive statistics computed directly from all 1,970 predicted WI values, including mean, median, standard deviation, wealth class percentages, WI histogram, DHS MAE, and province/municipality rankings).

The map renders all prediction points as a heatmap at low zoom and individual colored circles at high zoom, with province and municipality boundary highlights via

Mapbox feature-state. Selecting an area opens an analytics side panel presenting: predicted wealth index distribution with national benchmark, DHS Survey WI validation block, area summary statistics grid, service infrastructure donut chart, landuse composition donut chart, SHAP feature driver ranked list, and a prediction points list with DHS/grid source badges. Each area generates a downloadable PDF analytics report with nine sections: Area Overview, Point-Level Estimates, Wealth Distribution, Infrastructure Profile, Municipal Breakdown (province-level only), SHAP Feature Attribution, Policy Recommendations, National Comparison, and Methodology Summary.

4.6.1 Expert Acceptance and Evaluation

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